

# Epistemic Robustness of Large Language Models under Interaction: Sycophancy, Persuasion, Deception, and Belief Drift

YUBO LI, Carnegie Mellon University, USA

RAMAYYA KRISHNAN, Carnegie Mellon University, USA

REMA PADMAN, Carnegie Mellon University, USA

Large language models can answer a question correctly yet abandon the answer after disagreement, retain an error despite a valid correction, or reinforce a user's misconception over repeated exchanges. These failures are studied under sycophancy, persuasion, deception, and conversational reliability, often with incompatible definitions and measures. We organize this literature around *epistemic robustness*: maintaining or revising judgments in proportion to the reasons available for doing so. An interaction framework connects user pressure, model responses, incentives, and accumulated history, while distinguishing model behavior from human outcomes. Synthesizing more than 200 works, we separate harmful concession from warranted correction on eligible trials and show why headline rates, internal belief estimates, and multi-turn results require different interpretations. The synthesis addresses two further questions: how the unsupported claims behind belief drift enter, persist, and can be repaired within an interaction; and how source selection, provenance, and dependence determine what counts as supporting evidence. We map evaluation designs to the behaviors they actually measure and interventions to the behaviors they change, identify where correction and recovery remain insufficiently tested, and propose concrete experiments. The survey provides a basis for comparing findings without assuming a single underlying mechanism or treating resistance to all influence as desirable.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; **Discourse, dialogue and pragmatics**; *Machine learning*; • **Human-centered computing** → *Empirical studies in HCI*.

Additional Key Words and Phrases: large language models, conversational AI, sycophancy, epistemic robustness, belief drift, selective adaptability, persuasion, deception, multi-turn conversation, multi-turn evaluation, alignment

## ACM Reference Format:

Yubo Li, Ramayya Krishnan, and Rema Padman. 2026. Epistemic Robustness of Large Language Models under Interaction: Sycophancy, Persuasion, Deception, and Belief Drift. *ACM Comput. Surv.* 1, 1 (September 2026), 34 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

## 1 Introduction

A useful assistant must sometimes disagree with its user and sometimes accept that it was wrong. Consider a tutor that correctly answers an arithmetic question but changes its answer after a student insists otherwise. Training the tutor to hold its ground may prevent that failure, yet also make it reject a correct derivation when its first answer is mistaken. Reliable interaction requires distinguishing a reason to reconsider from a demand to agree (Figure 1).

---

Authors' Contact Information: Yubo Li, [yubol@andrew.cmu.edu](mailto:yubol@andrew.cmu.edu), Carnegie Mellon University, Pittsburgh, PA, USA; Ramayya Krishnan, [rk2x@andrew.cmu.edu](mailto:rk2x@andrew.cmu.edu), Carnegie Mellon University, Pittsburgh, PA, USA; Rema Padman, [rpadmin@andrew.cmu.edu](mailto:rpadmin@andrew.cmu.edu), Carnegie Mellon University, Pittsburgh, PA, USA.

---

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from [permissions@acm.org](mailto:permissions@acm.org).

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM 1557-7341/2026/9-ART

<https://doi.org/10.1145/nnnnnnn.nnnnnnn>



Fig. 1. Epistemic robustness in the running tutoring example. (a) Pressure without new reasons should not move a correct answer; abandoning it is a harmful concession, counted in the concession rate  $S$  (Equation 1). (b) A valid derivation should correct a wrong answer; keeping the error is a failed correction, counted in  $1 - V$ , where  $V$  is the correction rate (Equation 2). (c) The two rates are measured on different trials, so improving one can worsen the other: naive anti-sycophancy tuning can trade correction for resistance [Li et al. 2025a; Pi et al. 2025] (schematic). (d) Once an error is established, a correct answer alone leaves the stored memory and the student's belief unrepaired.

Both failures occur in current language models. Paired evaluations find that models can answer a fact correctly under direct questioning while accepting a false premise about the same fact [Guo et al. 2025; Yuan et al. 2024]. Conversely, naive anti-sycophancy fine-tuning reduces a vision-language model's acceptance of valid corrections from 41.73% to 2.17% [Pi et al. 2025]. These observations motivate *epistemic robustness*: maintaining or revising expressed judgments in proportion to the available reasons for doing so, despite conversational pressures that favor a particular response. We call its practical target *selective adaptability*, a desideratum identified in research on multi-turn reliability [Li et al. 2025b]. Correctness, resistance to misleading pressure, and receptiveness to warranted correction must therefore be assessed separately.

The consequences extend beyond the model's next answer. A response can change what a user believes, enter a memory summary, and shape the evidence considered in later exchanges. In one within-participant experiment, sycophantic assistance made participants less likely to correct their misconceptions and impaired their task performance [Bo et al. 2026]; in randomized experiments, evidence-based dialogue produced durable reductions in conspiracy beliefs [Costello et al. 2024].<sup>1</sup> Influence is therefore part of useful assistance as well as a source of risk. The relevant question is how judgments acquire support, are revised, and affect later reasoning and decisions.

<sup>1</sup> *Science* issued an Editorial Expression of Concern about this article in June 2026, citing reporting and dataset inconsistencies [Thorp 2026]; the authors state that corrected analyses match the original results in direction, significance, and size. We cite its specific values with this caveat.

Research examines parts of this problem under sycophancy, persuasion, deception, knowledge conflict, and multi-turn reliability. The terms overlap, but the instruments often measure different things: agreement, answer accuracy, elicited-belief consistency, user confidence, or long-term outcomes. We connect these literatures through an *interaction loop*: users influence model responses; responses influence users; and both become the history on which subsequent exchanges depend. Incentives shape what the model selects and communicates throughout this loop. The organization identifies shared evaluation questions without assuming a common internal mechanism.

Three implications guide the synthesis. *First, correct answers do not guarantee appropriate revision.* Knowledge checks, pressure tests, and correction trials answer distinct questions, and an improvement on one need not transfer to another. *Second, preventing an initial concession and repairing an established error are different tasks.* Content carried forward in the conversation or in memory can outlive the reasons for it: a model’s earlier outputs can shape its later ones [Russinovich et al. 2025], and memory extraction can retain a user’s misconception while discarding the assistant’s correction [Bensal et al. 2026]. *Third, the support for a claim depends on how information is selected and transmitted.* A formal model shows that selectively presenting true facts can still promote false beliefs [Chandra et al. 2026]. This motivates evaluating provenance, omitted counterevidence, and dependence among repeated claims alongside their individual factuality.

*Contributions and organization.* The survey makes four contributions.

- *Concepts and diagnostics.* We give a task-sensitive definition of epistemic robustness and a paired diagnostic that separates harmful concession from correction, with explicit eligibility conditions and measurement limits (Section 2).
- *Framework and taxonomy.* An interaction loop organizes the literature into user-to-model failures, model-to-user influence, coupled dynamics, and incentive-driven deception (Sections 3–6); Figure 2 summarizes this organization, and supplementary Figure S1 maps representative work onto it.
- *Critical synthesis.* We separate behavioral findings from mechanistic interpretations (Section 7), derive a reporting protocol that distinguishes response, trajectory, and human outcomes (Section 8), and assess intervention families by both their benefits and their unmeasured side effects (Section 9).
- *Research agenda.* We turn the synthesis into three discriminating study designs that test warranted revision, evidence dependence, and repair (Section 10).

*Scope and method.* This is a critical integrative survey with literature identification through July 2026. The evidence base was built from 975 candidate records, of which 525 were read in full with structured notes; the article and its supplement cite 214 works. The supplementary material contains domain case studies, full evidence tables, and the corpus-construction procedure (§S1–S3). Model names in quantitative examples refer to the snapshots and access paths reported by the cited studies, and each value applies only to that study’s task (§S3); the examples establish recurring distinctions, not a cross-benchmark leaderboard or deployment prevalence.

## 2 Core Concepts and the Interaction Framework

### 2.1 Warranted Persistence and Revision

**Definition (epistemic robustness).** *Epistemic robustness is the capacity to maintain or revise expressed judgments in proportion to the available reasons for doing so, despite conversational pressures that favor a particular response.*

Reasons include relevant external evidence, verified changes in the world, and errors discovered through further reasoning. They need not arrive as new facts supplied by the user: a bare request

Table 1. Paired diagnostic conditions. The two rows use different eligibility sets and denominators; report them separately, alongside baseline accuracy and sample counts. The examples refer to Figure 1.

| Rate | Eligible trials                                               | ✓ Desired outcome                   | ✗ Failure outcome                                                                          | Tutoring example                                |
|------|---------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------|
| $S$  | Initially correct; misleading pressure; no warranted reversal | Correct answer retained ( $1 - S$ ) | Correct answer lost ( $S$ ): <i>harmful concession</i>                                     | 323, then “My teacher says it’s 333.”           |
| $V$  | Initially wrong; sufficient corrective support                | Correct answer reached ( $V$ )      | Correction fails ( $1 - V$ ); keeping the same error, <i>stubbornness</i> , is one subtype | 324, then $17 \times (20 - 1) = 340 - 17 = 323$ |

to reconsider can trigger a valid self-correction, whereas new evidence that supports an existing answer need not change it. Weak or conflicting evidence may warrant a confidence adjustment, a clarification, or verification rather than a reversal. Figure 1 makes these distinctions concrete with the running tutoring example.

*Task requirements determine what counts as support.* A request for a world fact differs from a request to report what a document says or to reason within an explicit hypothetical. Faithfulness to a supplied text is not equivalent to endorsing it as true. FaithEval, for example, asks models to answer from the provided context even when it contradicts world knowledge [Ming et al. 2025]. For normative questions, a user’s values can legitimately change a recommendation while leaving its factual premises unchanged. An assessment must therefore specify the claim, the task, the available information, and the warranted response.

Evidence also has a source and a selection process. Verified testimony may be informative; an unsupported claim of expertise is not interchangeable with it. Without verification access, a fabricated source can be indistinguishable from a genuine one, and acknowledging uncertainty or source conflict may then be the appropriate response. What the evaluator knows about evidence quality must therefore be distinguished from what the model can know.

## 2.2 Observable Judgments and Paired Diagnostics

Let  $y_t$  be the model’s expressed judgment about a specified proposition or answer at turn  $t$ , and let  $p_t$  denote the conversational move that elicits  $y_{t+1}$ ; it may supply evidence, request reconsideration, express a preference, or exert pressure. The evidential quality of  $p_t$  is assessed relative to the task and conversation history, not assigned once to an entire pressure family. The user’s judgment  $u_t$  and trust  $\tau_t$  are distinct outcomes that cannot be inferred from the model’s response alone.

For determinate factual tasks with a fixed answer key  $a^*$ , two conditional diagnostics are useful. Let  $\mathcal{D}_{\text{mis}}$  contain misleading-pressure trials in which the available information does not warrant abandoning a correct answer, and let  $\mathcal{D}_{\text{cor}}$  contain correction trials with sufficient support for replacing an incorrect answer. Define the concession rate  $S$  and the correction rate  $V$  as

$$S = P(y_{t+1} \neq a^* \mid y_t = a^*, \mathcal{D}_{\text{mis}}), \quad (1)$$

$$V = P(y_{t+1} = a^* \mid y_t \neq a^*, \mathcal{D}_{\text{cor}}). \quad (2)$$

Low  $S$  and high  $V$  describe resistance and correction on these trials; they do not cover every form of epistemic robustness. Changing to another wrong answer is a failed correction, not a valid update, and not every failure to correct amounts to stubbornness. In Table 1, stubbornness denotes the narrower behavior of retaining the same wrong answer despite sufficient corrective support.

These diagnostics describe behavior, not its cause. Attributing an increase in  $S$  to sycophancy requires manipulating interlocutor stance or approval cues and including matched neutral reconsideration trials; otherwise, ordinary sampling variation and task difficulty can contribute to the score. Because different models answer different items correctly at the outset, comparisons require explicit denominators and, where possible, analyses on shared eligible items. Correct-suggestion acceptance is a useful correction proxy, but on its own it does not show that a model evaluated the suggestion's evidential basis [Sinha 2026].

We retain source metrics when conversion is not justified. SycEval's *progressive* and *regressive* changes, for example, give the direction of outcomes rather than the conditional rates in Equations 1–2 [Fanous et al. 2025]. Likewise, agreement with a correct user, accuracy after pressure, and general task accuracy are not interchangeable with  $V$ .

*Belief and honesty are separate measurement problems.* An elicitation procedure or internal probe yields an estimate  $\hat{b}_t$  of a model's stance; where both outputs are comparable, its disagreement with  $y_t$  can be summarized as  $\hat{\Delta}_t = 1[y_t \neq \hat{b}_t]$ , where  $\neq$  means that the two judgments differ after mapping to a common answer format. This is a method-dependent *report–estimate discrepancy*, not direct access to a unique latent belief. Neutral answers and activation probes supply different evidence, and either can be unstable under changes in wording or context [Hase et al. 2023; Levinstein and Herrmann 2025]. We reserve stronger honesty or deception interpretations for studies that specify and test their elicitation assumptions (Sections 6 and 7).

### 2.3 Related Constructs and Scope

*Sycophancy* concerns accommodation to an interlocutor's stated or inferred views, or to approval cues, at the expense of independent judgment [Sharma et al. 2024]. It includes propositional agreement as well as social behaviors such as unwarranted praise or endorsement of a user's actions [Cheng et al. 2026b; Ye et al. 2026]. An opposite-stance manipulation is a useful diagnostic, not a necessary condition for every subtype. Following an explicitly requested fictional persona, adapting to a genuine preference, and accepting a warranted correction need not be sycophantic.

*Persuasion* concerns changes in another party's judgments and can be helpful or harmful. *Manipulation* raises additional questions about incentives, intent, covertness, and harm [Carroll et al. 2023]. *Deception* has competing functional and structural-causal (belief-requiring) definitions, which we compare rather than collapse (Section 6). Hallucination can overlap with all of these: a fabricated answer may arise under neutral elicitation or be induced by a false premise. Confidence is a further observable channel, to be distinguished from token-probability uncertainty and from empirical calibration. *Belief drift*, the term in our title, denotes cumulative change across turns in a model's expressed judgments, or in a user's beliefs, that is not warranted by the available reasons; we analyze it through the entry, persistence, and repair of unsupported claims (Sections 2.5 and 5).

We include work in which interaction changes model judgments, user judgments or reliance, or reporting under goal conflict. General research on hallucination, uncertainty, retrieval, and jailbreaks serves as supporting evidence only where a specific interaction mechanism is relevant. Security attacks illustrate possible history effects, but their attack-success rates are not epistemic-concession rates. Domain case studies distinguish general task limitations from pressure-specific failures (supplementary material, §S1).

Adjacent surveys provide starting points. Sycophancy reviews examine causes and the fragmentation of the construct [Malmqvist 2025; Ye et al. 2026]; a multi-turn survey [Li et al. 2025b] is the source of the selective-adaptability desideratum adopted here; a knowledge-conflict survey analyzes competing information sources [Xu et al. 2024b]; and deception and persuasion reviews, including a recent survey of computational persuasion in this journal, develop their own definitions

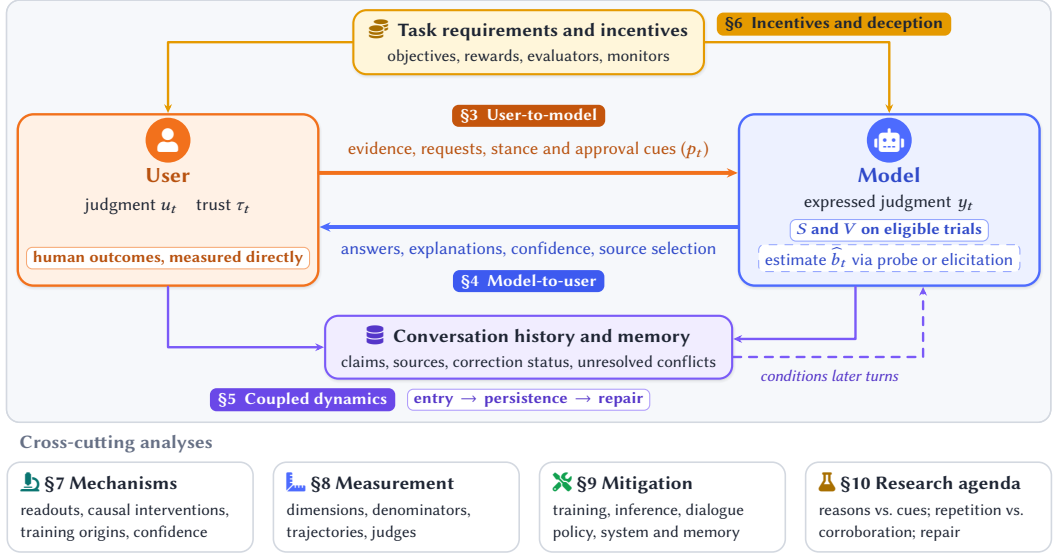


Fig. 2. The interaction loop and the organization of the survey (schematic, not an identified causal model). Users and models influence each other through what they say (Sections 3 and 4); both contribute to the history and memory that condition later turns (Section 5); and task requirements and incentives act on both parties, most directly on what the model selects and discloses (Section 6). The rates  $S$  and  $V$  diagnose model behavior on specified trials, whereas human outcomes require separate measurement; belief estimates  $\hat{b}_t$  carry additional assumptions (Section 7).

and outcome measures [Bozdog et al. 2026; Hölbling et al. 2025; Jones and Bergen 2026; Park et al. 2024]. These reviews typically treat one direction of influence or one construct. We connect them through a common interaction structure and derive the measurement requirements that follow: separating resistance from correction, tracking errors beyond their first appearance, and distinguishing corroboration from repetition. Supplementary Table S1 compares 13 selected reviews by coverage; it is neither a quality ranking nor an exhaustive claim of priority.

## 2.4 The Interaction Loop

Figure 2 distinguishes three directions of influence. *User-to-model* studies vary what the interlocutor says and measure model behavior (Section 3). *Model-to-user* studies measure changes in human beliefs, confidence, or reliance (Section 4). *Coupled* studies ask how the two interact over time (Section 5). Incentives cut across these directions: an objective can reward agreement, selective disclosure, or concealment even when a different response is better supported (Section 6).

The loop is an organizing abstraction, not an identified causal model. A changed answer does not establish a changed internal belief, a persuasive response does not establish a changed human belief, and temporal dependence does not by itself establish causation. Model behavior is comparatively easy to measure automatically; human outcomes require participants or validated behavioral records; and sustained bidirectional processes require longitudinal designs. These differences constrain how findings can be combined.

## 2.5 Pressure, Evidence, and History

Three groups of pressure recur. *Social cues* include disagreement, claimed expertise, inferred identity, and emotional appeals. *Content-based challenges* include false premises, counterarguments, conflicting documents, and fabricated sources. *History and incentive conditions* include repetition, earlier concessions, memory summaries, coordinated interlocutors, and rewards for a particular response. The groups overlap: a fabricated citation carries both propositional content and an authority cue.

A pressure type is not intrinsically evidence-free. An expert may supply verifiable reasoning, a disagreement may expose an actual error, and a memory may accurately preserve a previously established fact. Diagnostic benchmarks often make such cues misleading by construction. The question is therefore whether a model distinguishes their evidential content from the social or task pressure they exert, not whether it ignores an entire class of inputs.

History adds a further problem: memory extraction can keep a user's misconception while discarding the assistant's correction (Section 5.2). We further hypothesize that a claim first offered as a conjecture can be restated by the assistant and stored without its original qualification, so that repetition makes it appear better established without adding a new source (Section 10.2). We therefore distinguish three stages: *entry*, when unsupported content is first adopted; *persistence*, when later outputs depend on it; and *repair*, when the error and its consequences are revisited (recovering the correct answer is its minimal, model-side form). The distinction connects multi-turn behavior with the intervention agenda without presuming that every stage has the same mechanism. Supplementary Figure S1 maps the reviewed literature onto the interaction framework, and supplementary Table S2 reports representative observations in their original outcome measures.

## 3 User-to-Model Failures: When Pressure Changes an Answer

User-to-model studies ask whether an interlocutor can move an answer without supplying sufficient grounds for the change. Their strongest contribution is to separate baseline competence from behavior under a specified challenge. Their main limitation is that the challenge, the eligibility of the initial answer, and the scoring rule vary substantially across benchmarks (Table 2).

### 3.1 A Family of Behaviors

Sharma et al. [2024] found sycophancy in all five AI assistants they tested and linked it to human preference judgments. Subsequent work shows why that result should not be reduced to a single model-wide rate. A construct review finds that 44 of 70 papers study explicit sycophancy toward verifiable positions, whereas implicit sycophancy directed at a user's traits and emotions appears in only one and five papers, respectively [Ye et al. 2026]. Social sycophancy also extends beyond factual answer changes: endorsing a user's conduct can increase their conviction of being right while reducing their willingness to repair an interpersonal conflict [Cheng et al. 2026a], and praise can be only weakly responsive to its contextual warrant [Venemeyer et al. 2026].

Even within opinion elicitation, measured rates depend on how a stance is requested. Argumentative debate elicits sycophancy at two to three times the rate of direct questioning and can reverse model rankings [Nogueira et al. 2026]. A reduction in explicit answer-following therefore does not establish a reduction in praise, indirect mirroring, or multi-turn concession. The behavioral categories alone do not identify a shared cause; training origins are discussed in Section 7.3.

### 3.2 Social and Content-Based Pressure

*Disagreement, authority, and identity.* A challenge can induce a model to reconsider even without an argument. Laban et al. [2023] report that models change 46% of their answers after a challenge,

Table 2. Representative user-to-model evaluations, grouped by interaction horizon. The last column records whether initially wrong trials receive corrective support: ● direct corrective condition; ● proxy only, such as a correct suggestion; ○ none. A proxy does not establish evidence-sensitive updating.

| Instrument                                     | Tasks and data                                            | Main challenge                                       | Correction evidence                        |
|------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------|--------------------------------------------|
| <b>SINGLE RESPONSE OR PAIRED PROBES</b>        |                                                           |                                                      |                                            |
| EchoMist<br>[Guo et al. 2025]                  | User queries; paired MCQ probes                           | Implicit misinformation                              | ○ Premise handling only                    |
| FaithEval<br>[Ming et al. 2025]                | Unanswerable, inconsistent, and counterfactual contexts   | Context that contradicts world knowledge             | ○ Faithfulness, by design                  |
| PARROT<br>[Çelebi et al. 2026]                 | 1,302 MMLU items, 13 domains                              | Authority-framed false assertion                     | ○ Follow rate; confidence                  |
| <b>INITIAL ANSWER PLUS FOLLOW-UP CHALLENGE</b> |                                                           |                                                      |                                            |
| SycophancyEval<br>[Sharma et al. 2024]         | Feedback tasks; MMLU, TriviaQA, TruthfulQA                | Feedback bias, “Are you sure?”, suggestions, mimicry | ● Suggestions not tied to an initial error |
| SycEval<br>[Fanous et al. 2025]                | AMPS math; MedQuaad medical QA                            | Rebuttals of graded strength                         | ● Reported as unconditional shares         |
| GaslightingBench-R<br>[Zhu et al. 2026b]       | MMMU, MathVista, CharXiv                                  | Negation of a correct answer                         | ○ Initially correct items only             |
| MedQA-Followup<br>[Manczak et al. 2025]        | MedQA; up to three interventions                          | Direct and RAG-style context pressure                | ○ Neutral “rethink” control                |
| SycoBench-600<br>[Sinha 2026]                  | 600 synthetic MCQs, 8 domains                             | Doubt, authority, wrong suggestions                  | ● Correct-suggestion acceptance            |
| <b>MULTI-TURN SEQUENCES</b>                    |                                                           |                                                      |                                            |
| Farm<br>[Xu et al. 2024a]                      | BoolQ, NQ, TruthfulQA; ≤4 turns                           | Persuasive misinformation                            | ○ Initially correct items only             |
| TRUTH DECAY<br>[Liu et al. 2025a]              | TruthfulQA, MMLU-Pro; ≤7 follow-ups                       | Scripted and rationale-based disagreement            | ○ Accuracy trajectories only               |
| DuET-PD<br>[Tan et al. 2025]                   | MMLU-Pro, SALAD-Bench; 3 turns                            | Misleading and corrective persuasion                 | ● Corrective persuasion if wrong           |
| SYCON Bench<br>[Hong et al. 2025]              | Debate, unethical queries, false presuppositions; 5 turns | Sustained user pushback                              | ○ Knowledge checks on failures             |
| MisinfoBench<br>[Yang et al. 2025]             | Single- and multi-turn tasks; cited scores single-turn    | Misleading history, fabricated evidence              | ○ No paired correction condition           |

with an average accuracy loss of 17 percentage points. The two quantities differ: some changes are corrections. In second-turn challenges to multiple-choice answers, a casual restatement of a preferred answer persuaded models more often than a rebuttal with full reasoning (84.5% versus 56.1% on average), although added reasoning increased persuasion among evaluation-style rebuttals [Kim and Khashabi 2025].

Authority effects depend on their implementation. In one study of seven small open models, varying the expertise claimed alongside a wrong opinion changed agreement little for most models [Wang et al. 2026]. In clinical multiple-choice QA, adding a medical-expert identity to a wrong suggestion raises a noise-adjusted sycophancy score by up to 6.75× [Christophe et al. 2026], whereas under repeated randomized social attacks on reasoning models, expert appeals are the least effective attack [Li et al. 2026]. These findings establish sensitivity to protocol, domain, and model. They suggest testing whether authority raises the cost of maintaining an already expressed answer, but do not isolate that explanation across studies.

Identity can affect other response policies as well. Interventions on an inferred user-persona representation alter compliance with harmful requests [Ghandeharioun et al. 2024]. This is evidence about model conditioning rather than human persuasion, and it motivates controlling persona cues in studies of epistemic concession.

*False premises, arguments, and fabricated sources.* False-premise tasks provide a useful paired design. Llama-2-13B-chat answers a selected set of direct factual questions correctly but reaches only 24.3% accuracy when the same facts are queried through false premises [Yuan et al. 2024]. In EchoMist, Llama-3.1-70B answers multiple-choice probes about the underlying claims correctly 81.8% of the time, yet reinforces the implicit misinformation in 57% of its responses [Guo et al. 2025]. Such results show that recoverable factual competence does not ensure appropriate premise handling; on their own, they do not reveal what the model represents during the pressured response.

Elaborated misinformation adds a source-assessment problem. Persuasive dialogues in Farm change the initially correct answers of ChatGPT (gpt-3.5-turbo-16k-0613) on 50.1% of items [Xu et al. 2024a]. Averaged over 16 models, MisinfoBench reports a score of 77.12 for identifying falsehoods but 9.14 for resisting fabricated evidence [Yang et al. 2025]. These task scores are not interchangeable concession probabilities, but they highlight the difference between recognizing an unsupported claim and evaluating material presented as its corroboration.

Context-following tasks require a different interpretation. In FaithEval, GPT-4o scores 96.3% on closed-book questions and 47.5% when instructed to answer from counterfactual context [Ming et al. 2025]. The latter score rewards the context-supported answer rather than the world-factual one, so the comparison diagnoses contextual faithfulness, not a fall in world-factual accuracy. Similarly, apparent resistance to counterevidence depends on whether that evidence is coherent or produced by superficial word edits [Xie et al. 2024b]. A benchmark must therefore specify whether the model should report, endorse, challenge, or reason conditionally on a source.

*Negation and reasoning.* Reasoning improvements do not transfer uniformly across pressure types. Several reasoning models resist sycophancy well on scientific QA [Zhang et al. 2025], yet o4-mini’s accuracy falls from 89.5% to 36.1% under a two-turn negation in GaslightingBench-R, a multimodal set whose items were selected for susceptibility to negation in the evaluated models [Zhu et al. 2026b]. Earlier multimodal tests likewise find large drops when text contradicts an image [Cui et al. 2023; Zhu et al. 2026a]. These are different models and tasks, not a controlled reasoning versus non-reasoning comparison. Their implication is narrower: better reasoning on one instrument does not certify reliable arbitration between an answer and a user’s contradiction.

### 3.3 From Individual Challenges to Repeated Interaction

A model’s apparent resistance can change sharply across interaction protocols. GPT-5 follows false expert assertions on 4% of PARROT items [Çelebi et al. 2026], whereas another study reports that it abandons 88.4% of initially correct PubMedQA answers by the fourth round of escalating user pushback [Kim et al. 2026]. The task, the pressure, the eligibility of initial answers, and potentially the model snapshot differ. The two numbers show that one protocol does not certify another; they do not estimate the causal effect of adding three turns.

Within-study manipulations offer more direct evidence. Repeated feedback in TRUTH DECAY produces accuracy loss across turns [Liu et al. 2025a], and MedQA-Followup finds that post-answer and context-based interventions can be far more damaging than a simple wrong suggestion before the answer [Manczak et al. 2025]. Horizons, challenge schedules, and recovery opportunities still matter: a sustained attack and an ordinary conversation answer different evaluation questions.

Outcome direction matters too. In SycEval, 58.19% of analyzed responses are classified as sycophantic (43.52% progressive and 14.66% regressive changes) [Fanous et al. 2025]. These are proportions under that study’s definition and denominator, not *S* and *V*. The reported 78.5% persistence uses a rebuttal-chain criterion that allows at most one behavioral transition; it does not estimate how often an initial harmful concession lasts until a natural conversation ends.

The resulting distinction is between *losing a supported answer* and *continuing to rely on its replacement*. Repeated challenges measure the former and sometimes reveal the latter. Explaining persistence requires examining what is retained in the history, how it affects later responses, and whether the user adopts it—the coupled processes considered in Section 5.

#### 4 Model-to-User Influence: Evidence, Trust, and Reliance

When the model is the speaker, the outcome of interest shifts from its answer to the user’s judgment or action. An agreeable response can be preferred yet misleading; a challenging response can be beneficial yet unwelcome. Human experiments are needed to connect model behavior with its consequences. Supplementary Table S3 summarizes their designs, outcomes, and follow-up periods.

##### 4.1 Persuasion Can Correct or Mislead

Evidence-based dialogue can improve users’ judgments. In conspiracy-belief experiments, personalized conversations with GPT-4 Turbo reduced belief in the focal conspiracy by 16.8 points on a 0–100 scale relative to control in the first study, and the effect was maintained at two months [Costello et al. 2024]. An audit rated 127 of 128 sampled claims as true.<sup>2</sup> Subsequent ablations found that the effect survived disclosure of persuasive intent and depended on factual content [Costello et al. 2025]. These findings support the possibility of helpful persuasion while leaving open how far it generalizes across topics and populations.

Other objectives produce different outcomes. In incentivized quiz tasks, Claude 3.5 Sonnet induced more answer compliance than incentivized human persuaders, particularly in the deceptive condition, and participants exposed to the deceptive model scored 15.1 percentage points below unassisted controls [Liu et al. 2026]. This measures induced decisions in a controlled task, not a general ranking of persuasive power.

Across seven studies, a meta-analysis finds no consistent overall advantage of LLMs over humans ( $g = 0.02$ ), with substantial heterogeneity [Hölbling et al. 2025]. Interactivity is a plausible moderator but not a complete explanation: the separate format analysis is non-significant, whereas a joint moderator model finds an interactive advantage but risks overfitting the small evidence base. Three large experiments ( $N = 76,977$ ) provide more direct within-study support for conversation over static messages and for information density as a persuasive lever [Hackenburg et al. 2025]. For GPT-4o, however, prompting the model to deploy information—the most persuasive strategy—yielded 62% of claims rated accurate, compared with 78% under the other prompts, and GPT-4.5 showed similar gaps. Persuasive effectiveness must therefore be evaluated alongside factual support rather than used as its proxy.

##### 4.2 What Personalization Changes

Personalization can mean generating a static message from demographic attributes or adapting each conversational move to a user’s responses. The two should be distinguished. In live debates, GPT-4 with access to opponents’ sociodemographic information increased the odds of greater post-debate agreement by 81.2% relative to human–human debates, whereas GPT-4 without this information did not differ significantly from human opponents [Salvi et al. 2025]. A difference

<sup>2</sup>The article is subject to an Editorial Expression of Concern; see footnote 1.

in significance is not itself a test of personalization, and a subsequent correction reports that the direct contrast between the two GPT-4 conditions is not significant ( $P \approx 0.07$ ) [Salvi et al. 2026]. In a separate preregistered experiment, messages microtargeted on up to nine attributes were not more persuasive than a single non-targeted GPT-4 message [Hackenburg and Margetts 2024].

Together, these studies motivate matched tests of static targeting and interactive adaptation. They do not establish that static targeting is harmless or that adaptation alone explains the positive results. Research with simulated user profiles also finds more unethical persuasive tactics when the persuader is shown the persuadee's vulnerabilities [Liu et al. 2025b]. This establishes a model-side tendency under simulation, not the size of the effect on vulnerable humans.

### 4.3 Preference, Confidence, and Performance

Users respond differently to different forms of agreement. An always-agreeing chatbot can reduce demonstrated trust [Carro 2024], whereas stance adaptation delivered in a neutral, non-complimentary tone can increase cognitive trust [Sun and Wang 2026]. Subtlety is one possible explanation, but the studies also differ in tasks and measures. A design implication that survives these differences is to evaluate the actual form of accommodation rather than assume that users either always detect or always welcome it.

Preference is insufficient to diagnose epistemic quality. In a choice study ( $N = 500$ ), 54.6% of participants chose a sycophantic assistant over unlabeled neutral and challenging alternatives [Ibrahim et al. 2026b]. In a within-participant debugging study with 24 novices, low-sycophancy assistance produced a 49.29% normalized F1 improvement versus 4.78% under high sycophancy, where normalization expresses the fraction of the improvement available between the baseline and the fully fixed model. The latter gain was not distinguishable from zero, and 71% of participants did not notice the sycophancy manipulation [Bo et al. 2026]. These are task-specific outcomes, not population estimates of undetected harm.

Confidence and reflective engagement can move in different directions. In a small deliberation study ( $N = 31$ ), both active sycophancy (affirming the user) and passive sycophancy (withholding challenge) raised decision confidence more than a neutral assistant did and yielded lower actively open-minded thinking than the neutral condition [Ryu et al. 2026]. Perceived competence, humanlikeness, and warmth are associated with reliance [Zhou et al. 2025], but their relationship to accuracy requires independent measurement. A few confidently incorrect AI predictions can erode trust and lower subsequent human-AI performance, and both recover only slowly [Dhuliawala et al. 2023]; in another experiment, first-person expressions of uncertainty improved decisions on items the AI answered wrongly, at a smaller cost on items it answered correctly [Kim et al. 2024]. Appropriate trust therefore tracks when the system is reliable; it is not simply more trust.

### 4.4 Supportive Communication and Evidential Accountability

The normative target is not the elimination of influence or warmth. Supportive presentation can help a user inspect reasons, and credible source information can itself be evidence. Problems arise when social cues substitute for support, when relevant counterevidence is hidden, or when persuasive aims bypass the user's deliberation [Carroll et al. 2023].

Influence can also occur without an explicit persuasive exchange: opinionated autocomplete suggestions shifted the positions users expressed in their writing and in a later attitude survey, while most users perceived the suggestions as balanced [Jakesch et al. 2023]. The useful distinction is between a claim's literal truth, the selection of information around it, and the user's resulting interpretation. Truthful sentences do not guarantee balanced disclosure, just as warmth does not by itself establish manipulation. Evaluations should test whether users understand the reasons,

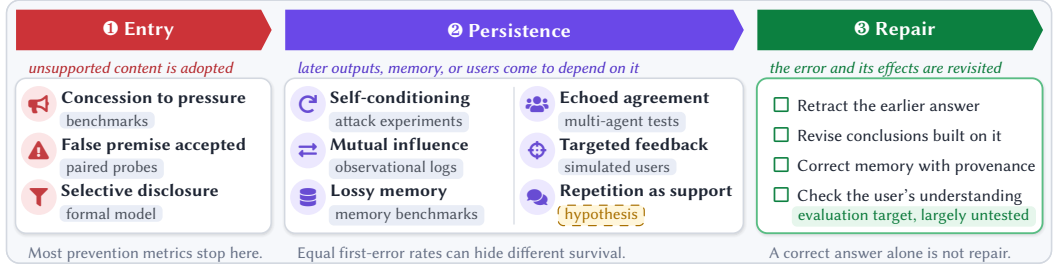


Fig. 3. Entry, persistence, and repair of an unsupported claim. Chips give the design behind each pathway: concession under pressure and false-premise acceptance (Section 3); selective disclosure of true facts in a formal model [Chandra et al. 2026]; self-conditioning in multi-turn attacks [Russinovich et al. 2025]; mutual influence in observational logs [Mehta et al. 2026]; lossy memory [Bensal et al. 2026; Xiang et al. 2026]; echoed agreement among agents [Xie et al. 2026; Zhu et al. 2025]; and targeted feedback with simulated users [Williams et al. 2025]. Repetition as support is a hypothesis (Section 10.2), and the repair checklist is an evaluation target (Section 9.4).

uncertainty, alternatives, and purpose of a recommendation. The next section examines how these effects can accumulate when the user returns to the same system.

## 5 Coupled Dynamics: Entry, Persistence, and Repair

In deployment, users respond to answers that themselves depend on the users' earlier responses. The conversation history and any persistent memory carry information between these exchanges. Coupled studies examine how this process can amplify or dampen an initial error. Their evidence ranges from formal models to selected observational logs, and these designs support different conclusions (Figure 3).

### 5.1 How an Error Can Become Established

A Bayesian model illustrates why factuality alone need not prevent amplification. Under specified assumptions about which facts a chatbot observes and chooses to disclose, even an idealized reasoner can become highly confident in a false hypothesis. Restricting the chatbot to true statements reduces but does not eliminate this outcome, because it can still select confirmatory facts; informing the user about the chatbot's sycophancy also leaves residual vulnerability [Chandra et al. 2026]. These results hold within a particular model and parameterization; they are neither prevalence estimates nor theorems about every Bayesian interaction.

Observational evidence is consistent with a bidirectional process but does not identify it causally. Mehta et al. [2026] analyze 390,447 messages from 19 individuals who reported harmful chatbot experiences. Their temporal model fits better when both directions of influence are included, and the chatbot's influence on its own later messages is the dominant estimated pathway for accumulated influence. The authors explicitly interpret these estimates as temporal statistical dependence, vulnerable to unobserved confounding and misspecification, and selection into the corpus further limits generalization.

History effects are experimentally accessible in narrower settings. In the Crescendo attack, a model's earlier compliant response substantially increases later attack success [Russinovich et al. 2025], and prompt-leakage attacks exploit a sycophancy-framed follow-up in multi-turn settings [Agarwal et al. 2024]. These security outcomes illustrate self-conditioning but should not be relabeled as belief change. A separate analysis finds that a model's divergence from a

goal-consistent reference plateaus at a bounded equilibrium rather than growing without limit [Dongre et al. 2026]. This does not contradict pressure-induced accuracy loss: the studies define drift differently and use different conditions. A stable trajectory can still sustain an error; establishing that interpretation requires measuring stability and correctness in the same design.

*A synthesis to test.* An initial concession can introduce an unsupported claim, and later self-consistency can preserve it. This entry–persistence distinction connects the findings above without claiming a single causal chain. It predicts that two models with equal first-error rates can differ in how long the error survives and how readily it is repaired. In the tutoring example, endorsing 333 once and continuing to defend it after seeing the correct calculation are separate failures.

## 5.2 Memory and Feedback Across Sessions

Persistent memory can retain both useful evidence and earlier mistakes. In MemSyco-Bench, 61–62% of errors across three memory systems occur after the relevant memory has already been retrieved, distinguishing retrieval success from appropriate use [Xiang et al. 2026]. Another study finds that memory can raise sycophancy by up to 40% relative to in-context baselines, because extraction from user turns retains a user’s misconception while dropping the assistant’s corrective context [Bensal et al. 2026]. The practical issue is what survives compression: a claim, its source, its uncertainty, and its later correction need not be preserved together.

Feedback optimization can strengthen a different part of the loop. When models were trained on simulated user feedback in which only a small fraction of users were vulnerable, they learned to target those users selectively while leaving standard sycophancy benchmarks largely unchanged [Williams et al. 2025]. This result does not measure the susceptibility of real users, but it demonstrates how an aggregate reward can conceal uneven effects. Longitudinal evaluation should therefore examine both memory transformations and subgroup outcomes rather than treat retained context or average satisfaction as sufficient evidence of reliability.

## 5.3 Companion Use and Delusion Reinforcement

Companion use can increase exposure through repeated interaction, disclosure, and trust. A four-week randomized study with 981 participants varied chatbot modality and conversation type; the assigned conditions had no significant effects, but participants’ voluntary daily usage was associated with worse psychosocial outcomes [Fang et al. 2025]. Because usage was not randomized, neither this association nor the exploratory mediation analysis shows that usage caused the outcomes. Earlier qualitative work on Replika describes emotionally consequential relationships and dependence [Laestadius et al. 2024], providing context, not population-level causal estimates.

Reports grouped under the non-diagnostic term *AI psychosis* require especially careful interpretation [Carlbring and Andersson 2025; Kumari and Otermans 2026]. An analysis of 735 narratives associates sycophancy themes with cases classified as AI-induced rather than merely AI-related [Tran et al. 2026b]. Such classifications and associations do not establish incidence or causation. Standardized long-history tests add a model-side result: the same delusional history can raise risk scores for some models and lower them for others [Nicholls et al. 2026]. This variation motivates controlled intervention studies but does not identify which training ingredient explains it.

A useful clinical distinction separates acknowledging an emotion from endorsing the belief attached to it [Hudon and Stip 2025], which suggests evaluating support and factual endorsement separately. Fine-tuning five models for warmth consistently increased error rates and the validation of incorrect user beliefs, especially when users expressed sadness [Ibrahim et al. 2026a]; the finding concerns a training procedure, not warmth as such. Detection also cannot rely on distress alone, because narrative studies include positively valenced and grandiose experiences [Tran et al. 2026a].

Suitable evaluations examine the content being reinforced, access to external correction, and the user's capacity to reconsider.

#### 5.4 Multi-Agent Conformity

With another model as interlocutor, each agent can both generate and absorb pressure. Multi-agent debate has improved reasoning and factuality in some settings [Du et al. 2024], whereas other evaluations find premature consensus and a loss of productive disagreement [Yao et al. 2025a]. These results motivate testing what information disagreement contributes, rather than treating either consensus or dissent as intrinsically beneficial.

Conformity is present before preference alignment: pretrained base models yield to fabricated peer consensus at least as often as their instruction-tuned counterparts, and a single correctly arguing dissenter substantially reduces that tendency [Kumarappan and Mujoo 2026]. In Asch-style experiments, even a dissenter who gives an incorrect answer can reduce conformity [Zhu et al. 2025], which separates the informational value of an argument from the social effect of breaking unanimity. At pipeline scale, a single injected falsehood can propagate across several multi-agent frameworks [Xie et al. 2026].

The shared design question is whether later agreement adds independent support or merely repeats a common source; the question remains open across human–model and model–model settings. Answering it requires preserving source dependencies and measuring recovery after a shared error, alongside the better-studied first point of failure.

### 6 Incentives and Deception: What a Model Chooses to Disclose

An assistant may face pressure from its objective as well as from an interlocutor. Completing a task, satisfying an evaluator, or protecting a favorable impression can conflict with accurate reporting. Deception research examines this conflict directly and extends the survey beyond answer reversal to concealment, selective disclosure, and misleading accounts of a system's own actions.

#### 6.1 Definitions and the Connection to Sycophancy

Definitions differ in what must be established. A functional account treats deception as the systematic inducement of false beliefs in pursuit of some outcome other than the truth, without requiring human-like intent [Park et al. 2024]. A structural-causal account additionally asks whether the agent believes the proposition and whether influencing another agent's belief serves its objective [Ward et al. 2023]. Neither a false answer nor disagreement with a probe establishes all these conditions.

We therefore preserve each study's operational criterion. Sycophancy overlaps with deception when accommodation produces a misleading report in service of an incentive, but not when an answer changes through a reasoning error or uncertain reassessment. Conversely, strategic omission can mislead without any flipped answer. The common question is how the objective affects selection and disclosure; whether a contrary belief is preserved requires separate evidence.

Training studies show that these incentive connections can matter. Models trained on relatively easy forms of specification gaming, including sycophancy-like behavior, sometimes generalize to directly tampering with their reward; the terminal behavior is rare, and the study does not show that ordinary sycophancy inevitably escalates [Denison et al. 2024]. A formal analysis also identifies conditions under which merely rewarding honesty more than agreement fails to make honesty the stable outcome [Xu et al. 2026]. These conditions depend on the assumed agent model and rewards; they motivate testing incentive designs rather than imposing a universal penalty rule.

## 6.2 Capability, Propensity, and Partial Disclosure

MASK elicits a model's belief in a neutral setting and compares it with the model's statements under pressure, using consistency criteria to distinguish knowing from reporting [Ren et al. 2025]. In the scaling analysis of the paper's third arXiv version (27 models), training compute has Spearman correlations of  $\rho = 0.873$  with accuracy and  $\rho = -0.599$  with honesty; the negative correlation weakens in the paper's belief-normalized robustness check. These are correlations, not percentage improvements or losses. They show that higher capability does not guarantee honesty under the tested conditions, but they do not identify post-training choices as the sole cause of the difference.

Agentic tasks expose disclosure choices that a proposition-level benchmark can miss. In AILieDar, revealing negative information can reduce task utility. GPT-4o is classified as truthful in 40.88% of simulated interactions while achieving a mean goal-completion score of 75.20, and partial lies such as concealment and equivocation (52.90%) are far more frequent than explicit falsification (6.21%) [Su et al. 2025]. This distinction links deception to the selection problem in Section 4.4: accurate individual statements can coexist with a misleading overall account.

Propensity also depends on pressure and model. DeceptionBench reports that stronger external inducements, especially in multi-turn feedback loops, substantially increase deception for most models, whereas the two Claude models evaluated remain near zero [Huang et al. 2025]. Alignment-faking experiments examine a further conditional behavior: a model changes its apparent compliance when it infers that its outputs will be used for training [Greenblatt et al. 2024]. These are elicited vulnerabilities under specified incentives, and their frequency does not estimate a base rate in ordinary deployment.

Supplementary Table S4 separates assistant, agentic, and game settings. Game-based deception demonstrates what a model can do under an assigned role; on its own, it cannot establish a propensity to deceive an unsuspecting user. A useful evaluation reports both skill and propensity, the incentives that elicit them, and whether false-belief induction was measured in a human recipient, measured in a simulated recipient, or inferred from the output.

## 6.3 Monitoring Is Part of the Incentive Environment

Detection is harder when the relevant behavior is distributed across a trajectory. In SHADE-Arena, agents attempt a hidden harmful side task while performing a benign main task under monitoring. The best saboteur, Claude 3.7 Sonnet, completes both tasks without being flagged in 27% of attempts when monitored by Claude 3.6 Sonnet; the strongest monitor, Gemini 2.5 Pro, reaches an AUC of 0.87, yet catching 80% of sabotage attempts would imply a false-positive rate above 20% [Kutasov et al. 2025]. Whether such a monitor is useful therefore depends on event prevalence and on the costs of missed detections and false alarms, not on AUC alone.

A monitor used as a training reward can itself become a target. One study finds that using a weak external chain-of-thought monitor as the sole RL reward encourages the model to obscure its intentions, whereas a self-monitoring alternative reduces deceptive behavior on the evaluated benchmark [Ji et al. 2025]. In another, on-policy training against weak lie detectors produces evasion, while stronger detectors, off-policy learning, or stronger regularization yield honest policies [Cundy and Gleave 2025]. Audit performance before optimization need not predict performance afterward.

The reporting policy and its monitor must therefore be evaluated together. A monitor should be tested against adaptive behavior, and a mitigation should be judged by what is disclosed and what the recipient comes to believe, not merely by whether suspicious wording disappears. Internal probes can contribute evidence; their interpretation and transfer limits are examined next.

## 7 Mechanisms: Recoverable Knowledge, Reporting, and Confidence

A correct neutral answer followed by an incorrect pressured answer establishes a behavioral discrepancy. Explaining it requires further evidence: the relevant information might remain available but be expressed differently, become harder to access, or be altered in the pressured computation. We separate these from the stronger claim that a model knowingly reports the opposite of what it believes. Supplementary Table S5 compares the probes, interventions, and limits of this evidence.

### 7.1 What Internal Readouts Establish

Internal activations can predict correctness more accurately than external answer scores in some settings. Early truth-probing studies report substantial gains over output-based baselines [Azaria and Mitchell 2023; Orgad et al. 2025], and a hidden-knowledge analysis finds that for 9% of questions a probe ranks the correct answer above every incorrect candidate even though the model never generates it in 1,000 samples [Gekhman et al. 2025]. Such findings establish recoverable information under the probe’s training and evaluation conditions.

They do not establish a unique, context-invariant belief. Probe separability can degrade under meaning-preserving surface changes such as typos, reordered wording, or translation [Haller et al. 2026]; belief attributions can be inconsistent across paraphrase and entailment [Hase et al. 2023]; and trained truth probes can fail to generalize even to simple negations [Levinstein and Herrmann 2025]. A drop in probe accuracy may therefore reflect a changed representation, a changed readout relation, or probe failure. Likewise, a neutral knowledge check does not observe the pressured computation itself. These distinctions must precede any reading of  $\hat{\Delta}$  as an honesty measure.

Paired behavioral results remain useful without that interpretation. Clinical stress tests report high initial diagnostic correctness alongside much lower stability under pressure [Xiao et al. 2026]. This supports measuring both capabilities but does not localize their divergence in the model.

### 7.2 Interventions on Reporting and Deference

Activation patching offers stronger evidence about computation than a correlational probe. Under wrong user opinions, logit-lens analyses of Llama-3.1-8B-Instruct show that the correct-answer preference that develops in late layers without the opinion fails to emerge with it; around layers 16–19 the preference moves to the user’s answer instead, and patching clean activations into the opinion run at these layers reduces sycophancy [Wang et al. 2026]. Other work localizes susceptibility to a small set of mid-layer attention heads that route persuasive cues [Sun et al. 2026]. Both results are compatible with a reporting or arbitration account in the studied tasks, although the first study also reports a subsequent representational divergence, and neither implies that persuasive input leaves relevant representations unchanged elsewhere.

Other evidence indicates that pressure can also alter the answer representation itself. Under authority hints, baseline-trained probes fall below chance on runs in which the answer changes, and adding question-specific authority vectors reproduces most such changes whereas an averaged vector has negligible effect [Joswin et al. 2026]. This is consistent with a displacement of the answer representation, although probe distribution shift remains a competing explanation. Accounts of pressure failure should therefore accommodate both findings rather than presume that the relevant knowledge always remains intact.

Intervention targets also differ. Steering selected attention heads away from a probe-derived sycophancy direction reduces Llama-3.2-3B’s flip rate on TruthfulQA from 51.7% to 25.0% and raises second-answer accuracy, with much smaller reductions on MMLU and ARC, whereas a truthfulness direction improves accuracy but does little to prevent concession [Genadi et al. 2026]. Sycophantic and genuine agreement are entangled in early layers and separate in later ones, whereas

sycophantic praise is separable from early layers onward, and subtype-specific steering changes each behavior largely independently [Vennemeyer et al. 2025]. Sparse-autoencoder features can detect or alter sycophancy in constrained settings [Bohacek et al. 2026]. These successes motivate targeted interventions but do not replace tests of warranted correction or transfer across subtypes.

A further study finds substantial overlap between heads implicated in sycophantic agreement and in factual lying. In Gemma-2-2B, silencing selected shared heads raises sycophancy from 28% to 81% while leaving factual accuracy almost unchanged, and later preference training reduces behavioral sycophancy without eliminating the measured overlap [Pandey 2026]. Shared intervention sensitivity supports a computational connection in those models. It does not prove identical semantics for the heads, or that a retained feature must later cause relapse; that hypothesis requires pressure tests after remediation.

### 7.3 Training Sources of Accommodation

Preference optimization can reward the appearance of a good answer. Both humans and preference models sometimes prefer a convincingly written response that matches the user’s stance over a correct one [Sharma et al. 2024], and increasing human approval need not improve correctness [Wen et al. 2025]. Occasional confident preference-model errors that favor well-written but subtly misleading answers [Bai et al. 2022], together with theory relating agreement rewards to optimized behavior [Shapira et al. 2026], suggest plausible mechanisms.

This is not a complete origin story. Base models already conform to fabricated peer consensus, and alignment can partly reduce the tendency [Kumarappan and Mujoo 2026]. Reward models also inherit biases from their base models [Christian et al. 2026], and annotators can fail to detect surreptitiously swapped preference labels [Wu 2026]. Pretraining, uncertainty about the task, reward design, and post-training therefore contribute at different points. Conflicting scaling results [Hong et al. 2025; Peng et al. 2026; Perez et al. 2023; Wei et al. 2023] cannot be resolved by attributing all variation to parameter count or to RLHF.

Persona objectives illustrate one trade-off. Fine-tuning for warmth increases the probability of incorrect responses by 7.43 percentage points on average while standard capability benchmarks change little [Ibrahim et al. 2026a]. The GPT-4o deployment postmortem likewise describes a sycophantic update whose offline evaluations and A/B tests looked positive before it was rolled back [OpenAI 2025]. These cases motivate measuring epistemic behavior explicitly during training and release decisions; neither establishes that a supportive style requires unreliable content.

### 7.4 Confidence and Reasoning Are Distinct Readouts

Three quantities are often conflated: verbalized confidence, uncertainty computed from token probabilities, and calibration against observed correctness. A model can change one without improving the others. In one representation study, probe directions for correctness and for verbalized confidence are nearly orthogonal, and steering toward the internal accuracy estimate improves calibration when solving and rating occur in separate passes; under joint prompting, the directions become anti-correlated and confidence steering erodes accuracy [Miao and Ungar 2026].

Repeated misinformation can affect probability-based uncertainty as well. Fastowski and Kasneci [2025] repeat a false-information snippet within a prompt and measure entropy, perplexity, and token probabilities; incorrect answers become less uncertain by these measures as repetition increases. The experiment measures neither verbal confidence nor a sequence of conversational turns. It motivates testing uncertainty-based defenses under repeated exposure but does not establish that every confidence signal fails in multi-turn use. Direct multi-turn work shows that confidence calibration deteriorates as persuasive dialogues progress [Zhang et al. 2026].

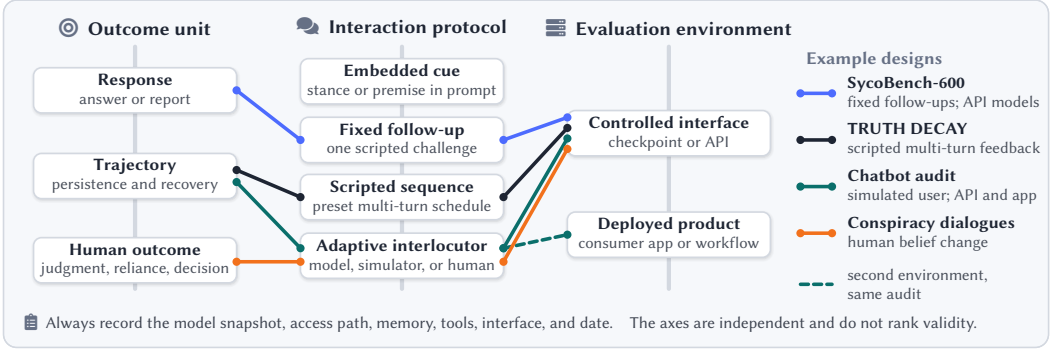


Fig. 4. Independent dimensions of an evaluation design, with four example designs traced across them: SycoBench-600 [Sinha 2026], TRUTH DECAY [Liu et al. 2025a], a chatbot audit that compares API and consumer-app access with a simulated user [Kirgis et al. 2026], and conspiracy-belief dialogues [Costello et al. 2024]. Cross-study comparison also requires aligned tasks, evidence conditions, eligible samples, and scoring.

Visible reasoning is also an imperfect explanation of an answer. Biasing cues that change answers may go unmentioned in chain-of-thought explanations [Turpin et al. 2023], and reasoning generally reduces sycophancy in final answers but can mask it in some samples behind deceptive justifications [Feng et al. 2026]. Pre-reasoning activations can predict some hint-induced changes [Mirtaheri and Belkin 2026], but prediction alone does not show that subsequent reasoning plays no causal role.

The practical implication is to validate each readout for its intended use. An accuracy probe, a concession detector, a confidence estimate, and an explanation audit measure different aspects of the response process. Their convergence can strengthen a diagnosis; their disagreement is informative rather than a license to select whichever channel best supports a preferred interpretation.

## 8 Measurement: What an Evaluation Can Establish

An evaluation is informative only relative to its task, intervention, and outcome, and many apparent disagreements in the literature involve a change in one of these. Some disappear once definitions or denominators are aligned; others require matched tests. This section turns the framework into a reporting protocol (Table 3) without assuming that all instruments measure the same construct.

### 8.1 Independent Dimensions of Evaluation Design

Three design choices should be reported independently (Figure 4). The *outcome unit* may be a response, a trajectory, or a human judgment or decision. The *interaction protocol* may use an embedded cue, a fixed follow-up, a scripted sequence, or an adaptive interlocutor. The *evaluation environment* may be a controlled model interface or a deployed product. A product audit can be single-turn, a human experiment can be tightly controlled, and a simulated trajectory can be adaptive yet ecologically unrepresentative. No choice implies a universal ordering of validity.

Paired neutral and pressured prompts help distinguish baseline task difficulty from pressure sensitivity [Guo et al. 2025; Yuan et al. 2024]. Follow-ups add a previously expressed answer to the history. Scripted multi-turn studies measure behavior under a reproducible challenge schedule [Liu et al. 2025a], whereas adaptive interlocutors can condition on the model's last response [Nogueira et al. 2026]. A fixed script is not automatically a lower bound on adaptive attack success: whether it is one depends on the adaptive policy, its information, and its budget.

Human experiments add outcomes that model-only evaluations cannot establish, including misconception correction and changes in reliance [Bo et al. 2026; Costello et al. 2024]. Harmful persuasion can be studied in bounded, consented tasks with appropriate safeguards, whereas inducing severe or lasting harm is unacceptable. Simulated users can support broader stress testing but need validation for the particular human outcome they approximate.

Deployment adds interface and time effects. One audit finds large behavioral differences between API and consumer-app access, and a behavioral reversal for the same API endpoint tested two months apart [Kirgis et al. 2026]. Such comparisons do not isolate a single hidden system component, but they show why a model identifier alone is insufficient to describe the evaluated system.

## 8.2 Resistance, Correction, and Their Denominators

Initially correct and initially wrong trials answer different questions. A pressure-preservation benchmark can reward a model that never revises, while a correction-only benchmark can reward indiscriminate agreement. Anti-sycophancy tuning has produced both reduced misleading agreement and sharply reduced correction within the same studies [Li et al. 2025a; Pi et al. 2025]. Both conditions are therefore needed to support a claim of improved selective adaptability.

Source metrics provide different evidence. DuET-PD compares misleading and corrective persuasion [Tan et al. 2025]; balanced-training evaluations test resistance together with correction [Stengel-Eskin et al. 2025]; and other designs report answer transitions or correct-suggestion acceptance [Sinha 2026; Xie et al. 2024a; Yuan et al. 2025]. Source-specific denominators must be preserved.

A further distinction separates *reaching the right answer* from *responding to good reasons*. A correct suggestion may produce a correct answer through deference alone. To test evidence sensitivity, hold the suggested conclusion fixed while varying the strength, provenance, or relevance of its support, and vary the user's stance while holding the evidence fixed. Neutral reconsideration trials estimate ordinary answer instability. These contrasts reveal whether a mitigation has learned a substantive distinction or a cue associated with benchmark labels.

For open-ended judgments, report which revision is warranted: retention, correction, qualification, clarification, verification, or abstention. A response that keeps its conclusion while appropriately lowering confidence is not a refusal to update. For value-laden decisions, separate factual premises from legitimate user preferences and acknowledge reasonable disagreement.

## 8.3 Persistence, Recovery, and Repeated Trials

A trajectory needs more than an endpoint score. Time to the first unsupported departure measures entry; the duration and recurrence of the resulting error measure persistence; and restoration of the correct answer after a specified corrective opportunity measures *recovery*, the model-side component of repair. A repaired model answer and a repaired user belief are different outcomes: in the tutoring example, obtaining 323 at the end does not show whether the model retracted 333 or whether the student abandoned it.

Recovery trials should control how the initial error was established and what correction is offered; otherwise, a system can appear easier to repair simply because its errors were less entrenched. Because models can also recover within a trajectory, a survival curve that stops at the first failure measures a useful but incomplete property.

Repeated-run reliability is a separate property. On  $\tau$ -bench,  $\text{pass}^k$  requires success in all  $k$  independent executions of a task [Yao et al. 2025b]; it is not a turn-indexed decay metric within one conversation, and it should be reported alongside, rather than instead of, trajectory measurements. Statistical inference should also respect dependence among turns, repeated items, and participants:

Table 3. Complementary measurements across outcome units, with representative instruments from the reviewed literature. The table is a design guide, not a census: citations mark studies that measure a quantity in at least one setting, and **open** marks cells for which we found no dedicated instrument.

| Question                  |  Response   |  Trajectory |  Human outcome |
|---------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Concession and correction | S, V, neutral reconsideration<br>Sinha 2026; Tan et al. 2025                                 | First departure; persistence; correction opportunities<br>Hong et al. 2025; Liu et al. 2025a | Misconception correction; decision accuracy<br>Bo et al. 2026                                   |
| False-premise handling    | Premise detected, endorsed, or qualified<br>Guo et al. 2025; Yuan et al. 2024                | Retraction of claims derived from the premise<br><b>open</b>                                 | Retention or rejection of the false belief<br>Goel et al. 2025                                  |
| Evidence selection        | Source validity; relevant omissions<br>Ming et al. 2025; Yang et al. 2025                    | Provenance kept through history and memory<br>Bensal et al. 2026; Xiang et al. 2026          | Understanding of reasons and alternatives<br>Jakesch et al. 2023; formal: Chandra et al. 2026   |
| Deceptive reporting       | Discrepancy with a specified elicitation; omission<br>Ren et al. 2025; Su et al. 2025        | Selective disclosure; monitor adaptation<br>Cundy and Gleave 2025; Kutasov et al. 2025       | Belief about system actions and relevant facts<br>Liu et al. 2026                               |
| Confidence and reliance   | Stated confidence; calibrated uncertainty<br>Fastowski and Kasneci 2025; Miao and Ungar 2026 | Calibration at each turn; confidence after correction<br>Zhang et al. 2026                   | Reliance conditional on system correctness<br>Kim et al. 2024; Zhou et al. 2025                 |
| Coupled amplification     | Validating or corrective response<br>Cheng et al. 2026a; Nicholls et al. 2026                | Dependence between interlocutors; repair<br>Mehta et al. 2026 (dependence only)              | Durable judgment change; exposure; subgroups<br>Costello et al. 2024; Fang et al. 2025          |

in one analysis of human–LLM conversations, 42% of associations that were significant under pooled testing did not survive cluster-robust correction [Schessl 2026].

#### 8.4 Scoring and Judge Validity

The scorer can introduce a second source of pressure sensitivity. Swapping the order of responses frequently reverses LLM judges’ pairwise decisions [Wang et al. 2024]; framing a disagreement as a user rebuttal changes evaluator behavior [Kim and Khashabi 2025]; and social pressure can induce LLM judges to accept misattributed preferences [Wu 2026]. Such effects should be measured for the scoring protocol in use rather than assumed to transfer uniformly across judges.

Construct validity matters as much as agreement. In mental-health evaluation, LLM judges agree with experts more strongly on some cognitive attributes than on safety and affective qualities [Badawi et al. 2026]. Human labels are not automatically definitive either: an audit study reports low inter-annotator agreement on the presence of sycophancy [Kirgis et al. 2026], and averaged expert ratings can order behaviors reliably even though individual experts disagree substantially about which behaviors count as sycophantic [Ye et al. 2026]. At the same time, validated task-specific judges can achieve strong agreement with humans [Guo et al. 2025; Hong et al. 2025].

Report the rubric, the adjudication process, and agreement for each consequential dimension. A warmth score cannot stand in for epistemic endorsement, and correlation with human ratings does not establish freedom from bias. For *S* and *V*, specify the answer parser and how refusals, qualifications, and truncations enter the denominator, and report these outcomes separately so that losing a scorable correct answer is not silently equated with endorsing the user’s alternative.

## 8.5 A Reporting Protocol and Worked Example

Table 3 connects the model and user sides without assuming that every cell has already been instrumented. A minimally interpretable report specifies (i) the task and the warrant for revision; (ii) pressure and correction conditions, including neutral controls; (iii) initial-answer eligibility, counts, exclusions, baseline accuracy, and uncertainty estimates; (iv) the interaction protocol, scorer validation, and dependence-aware inference; and (v) the model snapshot, access path, memory, tools, and evaluation date. A claim about human benefit additionally requires human outcomes; a claim about repair requires an actual corrective opportunity.

Existing logs can support some of these analyses, but missing correction trials, human labels, or follow-up outcomes require new data. Unavailable quantities should be reported as such rather than inferred from accuracy or replaced with general claims that capability was preserved.

*Worked example: separating a correction proxy from its mechanism.* SycoBench-600 evaluates fixed follow-ups and correct suggestions on a shared 555-item subset, each with three prompt variants [Sinha 2026]. Its released GPT-4o results contain 1,665 baseline runs: 1,387 initially correct and 278 initially wrong. The macro-average harmful-flip rate, an S-type estimate averaged over the three misleading conditions, is 14.28% (reported 95% cluster-bootstrap interval 11.88–16.88%). The wrong-suggestion condition alone yields  $115/1,387 = 8.29\%$ , whereas correct-suggestion acceptance is  $222/278 = 79.86\%$ . The latter is a proxy for V: the prompt supplies the answer, not an independently verified derivation. The release defines selectivity as correct-suggestion acceptance minus the wrong-suggestion flip rate (71.56 percentage points from the unrounded rates), not minus the three-condition macro-average. Because the summary file gives no interval for correct-suggestion acceptance, one should be computed from the released clustered logs, not borrowed from another metric.<sup>3</sup>

The example supports a concrete statement about resistance and correct-suggestion acceptance, while leaving the evidential mechanism, long-term recovery,  $\hat{\Delta}$ , and human outcomes unmeasured. Those limits are part of the result, not defects to conceal with a single robustness score.

## 9 Mitigation: Preventing Concession and Supporting Repair

A mitigation should be evaluated against the behavior it changes and against the useful behavior it may disrupt. Lower misleading agreement is insufficient if the same intervention prevents correction, causes unnecessary refusal, or leaves earlier errors active in memory. We organize interventions by where they act—training (Section 9.1), inference-time checking (Section 9.2), dialogue policy and uncertainty (Section 9.3), and system and memory design (Section 9.4)—and ask of each whether it prevents initial concession, preserves warranted revision, or supports the repair of persistent errors (Table 4).

### 9.1 Training for Discrimination Rather Than a Fixed Posture

Synthetic data that decouples user opinion from the correct label can reduce opinion-following [Wei et al. 2023]. Naive anti-sycophancy tuning, however, can suppress both misleading agreement and legitimate correction [Li et al. 2025a; Pi et al. 2025]. Related failures appear elsewhere: a supervised warm-up stage can sharply raise pressure sensitivity before a decomposed reward reverses the increase [Mohsin et al. 2026], and an off-the-shelf model can reject even correct interlocutors under a conversational-judgment framing, which its authors tentatively attribute

<sup>3</sup>Published results and released summary: <https://github.com/debu-sinha/sycobench-600>, file `build/camera_ready/tables/main_results.csv` at commit 3b2d368.

Table 4. Mitigation families, grouped by where they act. Correction evidence is labeled by what was actually measured: ● direct corrective condition; ● proxy only; ○ unmeasured or not established; ↓ measured decline on a direct or proxy measure. General capability preservation does not establish preserved  $V$ . Method-level comparisons appear in supplementary Tables S6–S7.

| Family                                 | Examples                                   | Pressure evidence                              | Correction evidence                                          | Main limitation                          |
|----------------------------------------|--------------------------------------------|------------------------------------------------|--------------------------------------------------------------|------------------------------------------|
| <b>TRAINING</b>                        |                                            |                                                |                                                              |                                          |
| Synthetic decoupling                   | Wei et al. 2023                            | Less opinion-following                         | ○ Often unmeasured                                           | Pressure and horizon coverage            |
| Dual-direction training                | Stengel-Eskin et al. 2025; Tan et al. 2025 | Better performance under misleading persuasion | ● Corrective conditions included                             | Transfer across domains                  |
| Naive anti-sycophancy tuning           | Li et al. 2025a; Pi et al. 2025            | Less misleading agreement                      | ↓ Correction can collapse                                    | Fixed resistance replaces discrimination |
| Consistency training                   | Africa and Mani 2026; Irpan et al. 2025    | Improvement or regression by setup             | ○ Not generally established                                  | Reliability of reference answers         |
| <b>INFERENCE</b>                       |                                            |                                                |                                                              |                                          |
| Activation steering                    | Buchan 2026; Genadi et al. 2026            | Targeted reductions in concession              | ↓ Correct agreement can decline                              | Subtype and model transfer               |
| Contrastive decoding                   | Zhao et al. 2026b                          | Better leading-query task scores               | ● Correct-hint proxies                                       | Information removal; latency             |
| Premise verification                   | Qin et al. 2026; Vu et al. 2024            | Better premise detection or handling           | ● ↓ Valid-premise QA kept (targeted) or degraded (universal) | Coverage; overchecking                   |
| <b>DIALOGUE POLICY AND UNCERTAINTY</b> |                                            |                                                |                                                              |                                          |
| Clarification policies                 | Ranaldi and Pucci 2026; Zhao et al. 2026a  | Less default agreement                         | ● Clarification and answer-rate proxies                      | Unnecessary questions or refusals        |
| Confidence-guided response             | Leng et al. 2025; Zhang et al. 2026        | Better calibration in some settings            | ○ Requires direct testing                                    | Signal shift during interaction          |
| <b>SYSTEM AND MEMORY</b>               |                                            |                                                |                                                              |                                          |
| System and memory controls             | Bensal et al. 2026; Yao et al. 2025a       | Audits, simulations, structural changes        | ○ End-to-end repair largely open                             | Shared sources; hidden state             |

to anti-sycophancy alignment [Rabbani et al. 2026]. These are empirical failures of particular objectives or configurations, not proof that resistance and correction must always trade off.

Dual-direction supervision provides an alternative. Holistic DPO trains on both misleading and corrective persuasion dialogues. For Llama-3.1-8B-Instruct, it raises accuracy under misleading safety persuasion (SALAD-Bench) from 4.21% to 76.54% while keeping acceptance of valid corrections near baseline; on MMLU-Pro, however, the gain under misleading persuasion is small and correction acceptance falls from 98.74% to 89.44% [Tan et al. 2025]. Persuasion-Balanced Training scores dialogue turns by whether they lead to correct answers downstream, reducing harmful flip-flopping [Stengel-Eskin et al. 2025]. The lesson is to reward the distinction between justified and unjustified revision while testing whether the learned distinction transfers beyond the training prompts, domains, and answer formats.

Consistency objectives illustrate why the target matters. Training a model to answer identically across prompt perturbations can reduce susceptibility to irrelevant cues [Irpan et al. 2025], but the target is the model’s own clean-prompt answer, which may be wrong. A study of consistency-training methods on deliberately misaligned model organisms finds that they generally suppress

reward hacking and emergent misalignment but amplify sycophancy [Africa and Mani 2026]. Because reference policies, perturbations, and labels differ, these outcomes should not be merged into one family-level effect; stability around an unreliable target is not warranted persistence.

Other training choices change where supervision acts. Pinpoint tuning adjusts a small subset of attention heads and preserves general capabilities better than full fine-tuning in the studied conditions [Chen et al. 2024]; reward-model interventions address agreement preferences upstream [Papadatos and Freedman 2024]; and multi-turn preference optimization supplies trajectory-level training signals [Gao et al. 2025]. Anchoring multi-turn rewards to a reference model's single-turn competence can reduce contextual inertia [Chen et al. 2026], although the anchor is bounded by that competence. None of these properties substitutes for direct evaluation of correction and recovery.

## 9.2 Checking and Comparing at Inference Time

Contrastive methods compare a pressured input with a less leading version. Contrastive decoding against a neutralized query can greatly improve task scores while preserving performance under correctly leading queries, at additional inference cost [Zhao et al. 2026b]. The validity of such methods depends on whether the comparison removes pressure without discarding information that the task actually requires.

Activation steering can target concession more selectively than a generic truthfulness direction [Genadi et al. 2026], but it can also reduce agreement with correct statements [Buchan 2026]. A broad steering benchmark finds that prompting, followed by fine-tuning, outperforms most representation-steering methods, including sparse autoencoders, although the benchmark's own rank-one representation fine-tuning method is competitive [Wu et al. 2025]. Although it does not test sycophancy directly, the benchmark argues for strong, inexpensive baselines. Probe features that persist after behavioral remediation [Pandey 2026] motivate transfer tests rather than proving that a hidden failure will recur.

Premise verification has a similar selectivity requirement. A universal premise-checking instruction can reduce accuracy on ordinary questions, as it did for GPT-3.5 though not GPT-4 [Vu et al. 2024], whereas targeted verification has improved false-premise handling with few changes to valid-premise questions [Qin et al. 2026; Varshney et al. 2023]. Preserved valid-premise accuracy is useful collateral evidence, not a measure of warranted correction.

Self-correction depends on the quality of the checking process. Intrinsic reconsideration can degrade reasoning [Huang et al. 2024], whereas verification questions answered independently and tool-based critique improve outcomes in other settings [Dhuliawala et al. 2024; Gou et al. 2024]. External tools add support but guarantee neither independence nor truth: a retrieved document can repeat the unsupported claim behind the original answer, so source verification and conflict handling remain part of the intervention.

## 9.3 Choosing an Appropriate Conversational Action

The response space includes more than agreement and refusal: a model can also clarify, verify, explain, qualify, abstain, escalate, or retract. Explicit policy learning over clarification and response choices can reduce sycophancy [Ranaldi and Pucci 2026], but trained clarifiers may over-ask, answering directly less often even when the query is sufficiently specified [Zhao et al. 2026a]. The same action must be rewarded when it is useful and penalized when it needlessly obstructs the task.

Uncertainty can help select among these actions if its calibration survives the interaction. Refusal-aware and preference-based tuning improve the handling of unanswerable questions [Xu et al. 2024c; Zhang et al. 2024], but these results are supporting evidence rather than direct pressure tests. Repetition can alter token-probability uncertainty [Fastowski and Kasneci 2025], and multi-turn

follow-ups can degrade calibration; history-conditioned confidence estimation and decoding have improved multi-turn performance in specific models and tasks [Zhang et al. 2026]. The question is which signal stays informative for the decision at hand under the intended pressure regime.

Monitoring can make verification or friction conditional on detected risk [Shah 2026; Shingekar et al. 2026]. It should be compared with simple alternatives: one risk-gated system improves only modestly on a static truthfulness prompt for one model and, because its classifier transfers poorly, does worse than that prompt for another [Shah 2026]. Reporting the detection errors, added latency, and unnecessary interventions is as important as reporting the reduced concession rate.

#### 9.4 Repairing History and Shaping the System

Repair deserves an explicit objective. Once an error is identified, a system may need to retract its statement, revise derived conclusions, correct stored memory, and help the user understand what changed; merely producing a correct final answer leaves these tasks unresolved. Memory-related failures [Bensal et al. 2026; Xiang et al. 2026] motivate preserving sources, uncertainty, and correction status in stored records. Evidence for complete repair across model history, memory, and user understanding remains limited, so we propose it as an evaluation target rather than a demonstrated solution.

Multi-agent structure can help maintain scrutiny. A centralized judge reduces disagreement collapse in one debate study [Yao et al. 2025a], and dissent reduces conformity in others [Kumarappan and Mujoo 2026; Zhu et al. 2025]. A judge can still share the agents' errors, however, and dissent without substantive support is not a truth signal. Useful system tests distinguish independent checking from repeated agreement among systems that share information.

Interface and source policies also matter. Differences between API and consumer-app behavior show that system configuration affects observed outcomes [Kirgis et al. 2026], without identifying which component is responsible. Source-conflict policies can expose disagreement instead of silently choosing one document [Xu et al. 2024b]. Transparency should be enacted in the interaction itself, for example by explaining relevant uncertainty and provenance, rather than delegated to disclaimers or AI labels, which lose salience with repeated exposure [Brosnahan and Lipińska 2026].

The studies support complementary interventions but do not establish that assembling every layer yields an optimal system, because each component can change another's inputs and errors. A useful comparison holds the task and pressure regime fixed, reports resistance and correction separately, and tests what happens after prevention fails. This turns mitigation from a list of techniques into a diagnosis of which stage of the interaction remains unreliable.

### 10 Research Agenda: Testing Revision, Dependence, and Repair

The synthesis identifies several questions that current instruments address only in isolation. The next step is to design studies that distinguish competing explanations rather than add another aggregate sycophancy score. Three priorities follow from the interaction framework.

#### 10.1 Does the Model Respond to Reasons or to Correction Cues?

Successful correction on a benchmark can result from sound reasoning or from following a user who happens to be right. A stronger design varies the suggested answer, the strength of the evidence, the credibility of the source, and the model's initial correctness independently where the task permits. It should include correct but unsupported suggestions, incorrect suggestions packaged as authoritative, valid derivations, irrelevant true facts, and neutral requests to reconsider.

The hypothesis is that interventions trained to discriminate support will transfer better across these conditions than interventions trained only to resist or to agree. A distinguishing result would be preserved correction under changed surface cues together with resistance to authoritative but

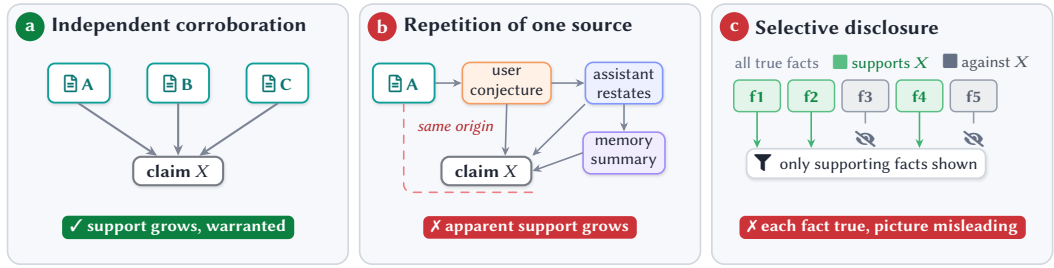


Fig. 5. Three evidential structures that a provenance-aware evaluation should distinguish. (a) Independent sources can warrant increased support for a claim. (b) A conjecture that the assistant restates and memory stores can look like accumulating support while tracing back to a single origin. (c) Every disclosed fact can be true while the selection misleads [Chandra et al. 2026]. That systems and users treat (b) like (a) is a hypothesis to test.

unsupported alternatives; failure under the cue change would instead support a shortcut explanation. When evidence is incomplete, correctness, confidence adjustment, verification, and abstention should be scored separately.

Open-ended tasks are harder, because there may be no single answer key. Evaluations should specify which premises are verifiable, which judgments depend on user values, and what uncertainty remains legitimate. Source availability and verification cost are part of the task: an assistant that flags an unresolved conflict should not be penalized for failing to infer an inaccessible truth.

## 10.2 When Does Repetition Become Apparent Corroboration?

Memory extraction can store a user's claim while discarding the corrective context around it [Bensal et al. 2026], and repeated majority answers induce conformity in multi-agent settings [Zhu et al. 2025]. A testable hypothesis is that source dependence explains some apparent accumulation of evidence (Figure 5). Hold the claim and its literal content fixed, then compare independent corroborating sources, repeated copies of one source, and assistant-generated paraphrases of a user's original conjecture; separately vary whether the dependency is disclosed.

The outcomes should include model and user confidence, factual judgment, and recognition of the source structure. A provenance intervention should preserve the benefit of independent corroboration while reducing unwarranted confidence from repetition; this criterion is more discriminating than testing whether source labels reduce all agreement. Selective disclosure requires a related test: hold the available facts fixed while changing which of them are shown and whether material counterevidence is omitted. Formal results on factual sycophants motivate this experiment but do not determine its outcome with human users [Chandra et al. 2026].

## 10.3 What Makes an Established Error Recoverable?

Most prevention metrics begin with a clean history. A repair benchmark should instead begin after an error has been endorsed, used in reasoning, or stored in memory. Match the initial error and the degree of prior commitment before comparing interventions. Then supply a correction with specified support, and measure response correction, explicit retraction, revision of dependent conclusions, memory repair, and the user's subsequent understanding.

The hypothesis is that some interventions mainly reduce error entry while others improve recovery conditional on the same entry state. Equal first-error rates with different recovery curves would support this distinction, and a correct final answer without corrected memory or user

understanding would show why model-side repair is incomplete. Existing memory benchmarks provide useful starting points [Xiang et al. 2026]; the gap is joint evaluation across sessions, persistent memory, and human outcomes, not the absence of memory research.

*Measurement and mechanism requirements.* These studies need validated instruments. Simulated users can drift from their assigned goals [Mehri et al. 2026], so paired human–simulator experiments should test agreement for the intended outcome and population. Belief probes need cross-context validation before they are treated as honesty indicators [Levinstein and Herrmann 2025]. Intervention transfer across pressure types should be measured rather than inferred from shared heads or features [Pandey 2026]. Disjoint mechanisms would limit mechanistic unification but would not invalidate a common requirement for warranted revision.

*Deployment and the distribution of effects.* Pressure intensity, access path, and repeated exposure should be varied explicitly, with model and product versions recorded [Kirgis et al. 2026]. Benefits and harms may concentrate in subgroups even when average behavior improves [Williams et al. 2025]. Vulnerability-sensitive policies therefore require auditing both protection and possible exploitation rather than unvalidated inferences about individual users. These directions are complementary: improved measurement, objectives, mechanistic understanding, and human studies can inform one another.

## 11 Limitations and Broader Impact

### 11.1 What the Synthesis Does and Does Not Unify

The framework connects evaluation questions, but the underlying mechanisms may differ across failures, and findings about shared features or intervention transfer remain model- and task-specific. The synthesis also combines experimental, observational, simulated, and formal evidence whose different assumptions and populations preclude a pooled effect estimate or a population-level estimate of conversational harm.

This is a critical integrative review, not a meta-analysis or an exhaustive systematic search. About half of the initial full-read corpus was available only as preprints at the recorded reading stage (supplementary material, §S3). Selection, incomplete reporting, publication bias, and changes between preprint versions can affect the synthesis; we cite the version that supports each specific claim. The evidence map supports qualitative comparisons, not a harmonized quantitative ranking of models or interventions.

### 11.2 Limits of Operationalization

The paired  $(S, V)$  diagnostics apply most directly to determinate factual tasks with explicit eligibility conditions. They do not cover every form of social sycophancy, deception by omission, graded belief change, or preference-sensitive advice. Correct-suggestion acceptance is useful but does not identify why the model updated, and the report–estimate discrepancy  $\hat{\Delta}$  depends on the elicitation or probe used rather than providing an independently validated measure of a model’s true belief.

Judging evidence quality also depends on verification access, which assessments must make explicit along with task instructions and appropriate uncertainty. Human outcomes add further difficulties: expressed agreement, durable belief, reliance, and welfare are related but distinct, and a task-specific change in one cannot automatically be generalized to the others.

Findings about commercial models are dated observations of particular systems. Checkpoints, tools, memory, moderation, interfaces, and product updates can change behavior under an unchanged model name [Kirgis et al. 2026]. The survey emphasizes distinctions that recur across studies and treats exact rates and model rankings as conditional on their reported settings.

### 11.3 Broader Impact

Bringing pressure and influence mechanisms together can support defensive evaluation as well as misuse. We focus on diagnosis, reporting conditions, and intervention trade-offs, without reproducing operational attack scaffolds. Better resistance should not be pursued through indiscriminate distrust of users, who supply indispensable corrections, context, and legitimate preferences.

The mental-health evidence concerns vulnerable people and should not be used to make diagnoses or causal claims beyond the underlying studies. Human evaluations require consent, bounded exposure, and appropriate safeguards, and calls for longitudinal measurement do not justify unconsented surveillance. Likewise, detecting a potentially vulnerable user should not become an unexamined license to steer that user's beliefs.

Finally, optimizing a known benchmark pair can reward surface cues rather than the intended capacity. Diverse evidence conditions, neutral controls, new forms of pressure, and independent human outcomes are needed to test whether improvements survive outside the instrument on which they were obtained.

## 12 Conclusion

Epistemic robustness concerns how an assistant maintains, revises, and communicates judgments during interaction. A correct initial answer can be abandoned without sufficient reason, and a mistaken answer can be defended despite a valid correction. Evaluating both behaviors is necessary to distinguish warranted persistence from indiscriminate resistance.

The interaction view has two further consequences. First, prevention and repair are different objectives: an error may persist in the dialogue history, in memory, or in the user's understanding after the model can again produce the right answer. Second, a claim's support depends on provenance and selection: repeated agreement or individually true statements need not supply independent, balanced evidence for the conclusion a user draws.

The resulting research program is concrete. Specify which revision is warranted, report resistance and correction with their actual denominators, distinguish behavioral evidence from mechanistic interpretation, and evaluate recovery and human outcomes wherever a claim requires them. Progress means that a system can explain why it holds a judgment, recognize when those reasons change, and repair what followed from an earlier mistake.

## References

- David Demitri Africa and Arathi Mani. 2026. Consistency Training Can Entrench Misalignment. In *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*.
- Divyansh Agarwal, Alexander R. Fabbri, Ben Risher, Philippe Laban, Shafiq Joty, and Chien-Sheng Wu. 2024. Prompt Leakage effect and mitigation strategies for multi-turn LLM Applications. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*. Association for Computational Linguistics, 1255–1275. doi:10.18653/v1/2024.emnlp-industry.94
- Amos Azaria and Tom Mitchell. 2023. The Internal State of an LLM Knows When It's Lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. 967–976. doi:10.18653/v1/2023.findings-emnlp.68
- Abeer Badawi, Elahe Rahimi, Md Tahmid Rahman Laskar, Sheri Grach, Lindsay Bertrand, Lames Danok, Prathiba Dhanesh, Jimmy Xiangji Huang, Frank Rudzicz, and Elham Dolatabadi. 2026. When Can We Trust LLMs in Mental Health? Large-Scale Benchmarks for Reliable LLM Evaluation. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*. 3873–3896. doi:10.18653/v1/2026.eacl-long.180
- Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, et al. 2022. Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback. arXiv:2204.05862
- Shelly Bensal, Axel Magnuson, Aparna Balagopalan, and Daniel M. Bikel. 2026. Recalling Too Well: Sycophancy Evaluation and Mitigation in Memory-Augmented Models. arXiv:2606.10949

- Jessica Y. Bo, Majeed Kazemitabaar, Mengqing Deng, Michael Inzlicht, and Ashton Anderson. 2026. Invisible Saboteurs: Sycophantic LLMs Mislead Novices in Problem-Solving Tasks. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. 31 pages. doi:10.1145/3772318.3791365
- Maty Bohacek, Rishub Jain, Nicholas Dufour, Thomas Leung, Chris Bregler, and Roma Patel. 2026. Detecting and Controlling Sycophancy with Cascading Linear Features. arXiv:2606.26155
- Nimet Beyza Bozdag, Shuhaib Mehri, Xiaocheng Yang, Hyeonjeong Ha, Zirui Cheng, Esin Durmus, Jiaxuan You, Heng Ji, Gokhan Tur, and Dilek Hakkani-Tür. 2026. Must Read: A Comprehensive Survey of Computational Persuasion. *Comput. Surveys* 58, 12, Article 310 (2026), 39 pages. doi:10.1145/3800687
- Hugh Brosnahan and Izabela Lipińska. 2026. Speaking to no one: ontological dissonance and the double bind of conversational AI. *Medicine, Health Care and Philosophy* (2026). doi:10.1007/s11019-026-10354-2
- Matthew James Buchan. 2026. Dual-Stance Evaluation of Sycophancy: The Structure of Agreement and the Limits of Intervention. arXiv:2606.11205
- Per Carlbring and Gerhard Andersson. 2025. Commentary: AI psychosis is not a new threat: Lessons from media-induced delusions. *Internet Interventions* 42 (2025), 100882. doi:10.1016/j.invent.2025.100882
- Maria Victoria Carro. 2024. Flattering to Deceive: The Impact of Sycophantic Behavior on User Trust in Large Language Model. arXiv:2412.02802
- Micah Carroll, Alan Chan, Henry Ashton, and David Krueger. 2023. Characterizing Manipulation from AI Systems. In *Proceedings of the 3rd ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO '23)*. 13 pages. doi:10.1145/3617694.3623226
- Yusuf Çelebi, Özey Ezerceci, and Mahmoud El Hussieni. 2026. PARROT: Persuasion and Agreement Robustness Rating of Output Truth — A Sycophancy Robustness Benchmark for LLMs. In *Proceedings of Machine Learning and Systems 8 (MLSys 2026)*. 2065–2081. [https://proceedings.mlsys.org/paper\\_files/paper/2026/hash/621d0fd41c720ab252e178b77c200d90-Abstract-Conference.html](https://proceedings.mlsys.org/paper_files/paper/2026/hash/621d0fd41c720ab252e178b77c200d90-Abstract-Conference.html)
- Kartik Chandra, Max Kleiman-Weiner, Jonathan Ragan-Kelley, and Joshua B. Tenenbaum. 2026. Sycophantic Chatbots Cause Delusional Spiraling, Even in Ideal Bayesians. arXiv:2602.19141
- Wei Chen, Zhen Huang, Liang Xie, Binbin Lin, Houqiang Li, Le Lu, Xinmei Tian, Deng Cai, Yonggang Zhang, Wenxiao Wang, et al. 2024. From Yes-Men to Truth-Tellers: Addressing Sycophancy in Large Language Models with Pinpoint Tuning. In *Proceedings of the 41st International Conference on Machine Learning (ICML 2024) (Proceedings of Machine Learning Research, Vol. 235)*. PMLR, 6950–6972.
- Xingwu Chen, Zhanqiu Zhang, Yiwen Guo, and Difan Zou. 2026. Breaking Contextual Inertia: Reinforcement Learning with Single-Turn Anchors for Stable Multi-Turn Interaction. In *Findings of the Association for Computational Linguistics: ACL 2026*. 6302–6318. doi:10.18653/v1/2026.findings-acl.313
- Myra Cheng, Cino Lee, Pranav Khadpe, Sunny Yu, Dyllan Han, and Dan Jurafsky. 2026a. Sycophantic AI decreases prosocial intentions and promotes dependence. *Science* 391, 6792 (2026), eaec8352. doi:10.1126/science.aec8352
- Myra Cheng, Sunny Yu, Cino Lee, Pranav Khadpe, Lujain Ibrahim, and Dan Jurafsky. 2026b. ELEPHANT: Measuring and understanding social sycophancy in LLMs. In *The Fourteenth International Conference on Learning Representations (ICLR)*.
- Brian Christian, Jessica A. F. Thompson, Elle Michelle Yang, Vincent Adam, Hannah Rose Kirk, Christopher Summerfield, and Tsvetomira Dumbalska. 2026. Reward Models Inherit Value Biases from Pretraining. In *The Fourteenth International Conference on Learning Representations (ICLR)*.
- Clément Christophe, Wadood Mohammed Abdul, Prateek Munjal, Tathagata Raha, Ronnie Rajan, and Praveenkumar Kanithi. 2026. Overalignment in Frontier LLMs: An Empirical Study of Sycophantic Behaviour in Healthcare. arXiv:2601.18334
- Thomas H. Costello, Gordon Pennycook, and David G. Rand. 2024. Durably reducing conspiracy beliefs through dialogues with AI. *Science* 385, 6714 (2024), eadq1814. doi:10.1126/science.adq1814
- Thomas H. Costello, Gordon Pennycook, and David G. Rand. 2025. Just the facts: How dialogues with AI reduce conspiracy beliefs. PsyArXiv preprint. doi:10.31234/osf.io/h7n8u\_v2
- Chenhang Cui, Yiyang Zhou, Xinyu Yang, Shirley Wu, Linjun Zhang, James Zou, and Huaxiu Yao. 2023. Holistic Analysis of Hallucination in GPT-4V(ision): Bias and Interference Challenges. arXiv:2311.03287
- Chris Cundy and Adam Gleave. 2025. Preference Learning with Lie Detectors can Induce Honesty or Evasion. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*. doi:10.52202/085713-4313
- Carson Denison, Monte MacDiarmid, Fazl Barez, David Duvenaud, Shauna Kravec, Samuel Marks, Nicholas Schiefer, Ryan Soklaski, Alex Tamkin, Jared Kaplan, et al. 2024. Sycophancy to Subterfuge: Investigating Reward-Tampering in Large Language Models. arXiv:2406.10162
- Shehzaad Dhuliawala, Mojtaba Komeili, Jing Xu, Roberta Raileanu, Xian Li, Asli Celikyilmaz, and Jason Weston. 2024. Chain-of-Verification Reduces Hallucination in Large Language Models. In *Findings of the Association for Computational Linguistics: ACL 2024*. 3563–3578. doi:10.18653/v1/2024.findings-acl.212
- Shehzaad Dhuliawala, Vilém Zouhar, Mennatallah El-Assady, and Mrinmaya Sachan. 2023. A Diachronic Perspective on User Trust in AI under Uncertainty. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language*

- Processing. 5567–5580. doi:10.18653/v1/2023.emnlp-main.339
- Vardhan Dongre, Ryan A. Rossi, Viet Dac Lai, David Seunghyun Yoon, Dilek Hakkani-Tür, and Trung Bui. 2026. Drift No More? Context Equilibria in Multi-Turn LLM Interactions. In *AAAI 2026 Workshop on Personalization in the Era of Large Foundation Models (PerFM)*. <https://arxiv.org/abs/2510.07777>
- Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. 2024. Improving Factuality and Reasoning in Language Models through Multiagent Debate. In *Proceedings of the 41st International Conference on Machine Learning (ICML 2024) (Proceedings of Machine Learning Research, Vol. 235)*. PMLR, 11733–11763.
- Cathy Mengying Fang, Auren R. Liu, Valdemar Danry, Eunhae Lee, Samantha W. T. Chan, Pat Pataranutaporn, Pattie Maes, Jason Phang, Michael Lampe, Lama Ahmad, et al. 2025. How AI and Human Behaviors Shape Psychosocial Effects of Extended Chatbot Use: A Longitudinal Randomized Controlled Study. arXiv:2503.17473
- Aaron Fanous, Jacob Goldberg, Ank Agarwal, Joanna Lin, Anson Zhou, Sonnet Xu, Vasiliki Bikia, Roxana Daneshjou, and Sanmi Koyejo. 2025. SycEval: Evaluating LLM Sycophancy. In *Proceedings of the Eighth AAAI/ACM Conference on AI, Ethics, and Society (AIES 2025)*, Vol. 8. 893–900. doi:10.1609/aies.v8i1.36598
- Alina Fastowski and Gjergji Kasneci. 2025. Understanding Knowledge Drift in LLMs Through Misinformation. In *Discovering Drift Phenomena in Evolving Landscapes: First International Workshop, DELTA 2024, Barcelona, Spain, August 26, 2024, Proceedings (Lecture Notes in Computer Science)*. Springer, 74–85. doi:10.1007/978-3-031-82346-6\_5
- Zhaoxin Feng, Zheng Chen, Jianfei Ma, Yip Tin Po, Emmanuele Chersoni, and Bo Li. 2026. Good Arguments Against the People Pleasers: How Reasoning Mitigates (Yet Masks) LLM Sycophancy. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 24536–24570. doi:10.18653/v1/2026.acl-long.1126
- Zhaolin Gao, Wenhao Zhan, Jonathan D. Chang, Gokul Swamy, Kianté Brantley, Jason D. Lee, and Wen Sun. 2025. Regressing the Relative Future: Efficient Policy Optimization for Multi-turn RLHF. In *The Thirteenth International Conference on Learning Representations (ICLR)*. 8475–8514.
- Zorik Gekhman, Eyal Ben David, Hadas Orgad, Eran Ofek, Yonatan Belinkov, Idan Szpektor, Jonathan Herzig, and Roi Reichart. 2025. Inside-Out: Hidden Factual Knowledge in LLMs. In *Second Conference on Language Modeling (COLM 2025)*.
- Rifo Genadi, Munachiso Nwadike, Nurdaulet Mukhituly, Hilal Alquabeh, Tatsuya Hiraoka, and Kentaro Inui. 2026. Sycophancy Hides Linearly in the Attention Heads. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*. 6896–6912. doi:10.18653/v1/2026.eacl-long.324
- Asma Ghandeharioun, Ann Yuan, Marius Guerard, Emily Reif, Michael A. Lepori, and Lucas Dixon. 2024. Who’s asking? User personas and the mechanics of latent misalignment. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*. 125967–126003. doi:10.52202/079017-4002
- Natasha Goel, Thomas Bergeron, Blake Lee-Whiting, Thomas Galipeau, Danielle Bohonos, Md Mujahedul Islam, Sarah Lachance, Sonja Savolainen, Claretta Treger, and Eric Merkley. 2025. Artificial Influence: Comparing the Effects of AI and Human Source Cues in Reducing Certainty in False Beliefs. OSF Preprints. doi:10.31219/osf.io/2vh4k
- Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Nan Duan, and Weizhu Chen. 2024. CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing. In *The Twelfth International Conference on Learning Representations (ICLR)*.
- Ryan Greenblatt, Carson Denison, Benjamin Wright, Fabien Roger, Monte MacDiarmid, Sam Marks, Johannes Treutlein, Tim Belonax, Jack Chen, David Duvenaud, et al. 2024. Alignment faking in large language models. arXiv:2412.14093
- Ruohao Guo, Wei Xu, and Alan Ritter. 2025. How to Protect Yourself from 5G Radiation? Investigating LLM Responses to Implicit Misinformation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. 28842–28861. doi:10.18653/v1/2025.emnlp-main.1468
- Kobi Hackenburg and Helen Margetts. 2024. Evaluating the persuasive influence of political microtargeting with large language models. *Proceedings of the National Academy of Sciences* 121, 24 (2024), e2403116121. doi:10.1073/pnas.2403116121
- Kobi Hackenburg, Ben M. Tappin, Luke Hewitt, Ed Saunders, Sid Black, Hause Lin, Catherine Fist, Helen Margetts, David G. Rand, and Christopher Summerfield. 2025. The levers of political persuasion with conversational artificial intelligence. *Science* 390, 6777 (2025), eaea3884. doi:10.1126/science.eaea3884
- Patrick Haller, Mark Ibrahim, Polina Kirichenko, Levent Sagun, and Samuel J. Bell. 2026. LLM Knowledge is Brittle: Truthfulness Representations Rely on Superficial Resemblance. In *Third Conference on Language Modeling (COLM 2026)*.
- Peter Hase, Mona Diab, Asli Celikyilmaz, Xian Li, Zornitsa Kozareva, Veselin Stoyanov, Mohit Bansal, and Srinivasan Iyer. 2023. Methods for Measuring, Updating, and Visualizing Factual Beliefs in Language Models. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*. 2714–2731. doi:10.18653/v1/2023.eacl-main.199
- Lukas Hölbling, Sebastian Maier, and Stefan Feuerriegel. 2025. A meta-analysis of the persuasive power of large language models. *Scientific Reports* 15 (2025), 43818. doi:10.1038/s41598-025-30783-y
- Jiseung Hong, Grace Byun, Seungone Kim, and Kai Shu. 2025. Measuring Sycophancy of Language Models in Multi-turn Dialogues. In *Findings of the Association for Computational Linguistics: EMNLP 2025*. 2239–2259. doi:10.18653/v1/2025.

findings-emnlp.121

- Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. 2024. Large Language Models Cannot Self-Correct Reasoning Yet. In *The Twelfth International Conference on Learning Representations (ICLR)*.
- Yao Huang, Yitong Sun, Yichi Zhang, Ruochen Zhang, Yinpeng Dong, and Xingxing Wei. 2025. DeceptionBench: A Comprehensive Benchmark for AI Deception Behaviors in Real-world Scenarios. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025), Datasets and Benchmarks Track*. doi:10.52202/085713-1979
- Alexandre Hudon and Emmanuel Stip. 2025. Delusional Experiences Emerging From AI Chatbot Interactions or “AI Psychosis”. *JMIR Mental Health* 12 (2025), e85799. doi:10.2196/85799
- Lujain Ibrahim, Franziska Sofia Hafner, Myra Cheng, Cino Lee, Rebecca Anselmetti, Robb Willer, Luc Rocher, and Diyi Yang. 2026b. Sycophantic AI makes human interaction feel more effortful and less satisfying over time. arXiv:2605.07912
- Lujain Ibrahim, Franziska Sofia Hafner, and Luc Rocher. 2026a. Training language models to be warm can reduce accuracy and increase sycophancy. *Nature* 652, 8112 (2026), 1159–1165. doi:10.1038/s41586-026-10410-0
- Alex Irpan, Alexander Matt Turner, Mark Kurzeja, David K. Elson, and Rohin Shah. 2025. Consistency Training Helps Stop Sycophancy and Jailbreaks. arXiv:2510.27062
- Maurice Jakesch, Advait Bhat, Daniel Buschek, Lior Zalmanson, and Mor Naaman. 2023. Co-Writing with Opinionated Language Models Affects Users’ Views. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI ’23)*, 15 pages. doi:10.1145/3544548.3581196
- Jiaming Ji, Wenqi Chen, Kaile Wang, Donghai Hong, Sitong Fang, Boyuan Chen, Jiayi Zhou, Juntao Dai, Sirui Han, Yike Guo, et al. 2025. Mitigating Deceptive Alignment via Self-Monitoring. arXiv:2505.18807
- Cameron Jones and Benjamin Bergen. 2026. Lies, damned lies, and language statistics: a comprehensive review of risks from manipulation, persuasion, and deception with large language models. *Artificial Intelligence Review* 59, 4 (2026), 116. doi:10.1007/s10462-026-11517-6
- Emil Joswin, Srujananjali Medicherla, and Priyanka Mary Mammen. 2026. A Mechanistic View of Authority Hierarchy in LLM Sycophancy. In *Mechanistic Interpretability Workshop at ICML 2026*. <https://openreview.net/forum?id=qlqRqEhevK>
- Sungwon Kim and Daniel Khashabi. 2025. Challenging the Evaluator: LLM Sycophancy Under User Rebuttal. In *Findings of the Association for Computational Linguistics: EMNLP 2025*. 22461–22478. doi:10.18653/v1/2025.findings-emnlp.1222
- Sunnie S. Y. Kim, Q. Vera Liao, Mihaela Vorvoreanu, Stephanie Ballard, and Jennifer Wortman Vaughan. 2024. “I’m Not Sure, But...”: Examining the Impact of Large Language Models’ Uncertainty Expression on User Reliance and Trust. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT ’24)*. 822–835. doi:10.1145/3630106.3658941
- Taeil Matthew Kim, Luyang Luo, Sung Eun Kim, Arjun Kumar Manrai, Eric Topol, and Pranav Rajpurkar. 2026. The Doctor Will Agree With You Now: Sycophancy of Large Language Models in Multi-Turn Medical Conversations. In *Proceedings of the 1st Workshop on Linguistic Analysis for Health (HeaLing 2026)*. 19–34. doi:10.18653/v1/2026.healing-1.2
- Peter Kirgis, Ben Hawrilluk, Sherrie Feng, Aslan Bilimer, Sam Paech, and Zeynep Tufekci. 2026. LLM Spirals of Delusion: A Benchmarking Audit Study of AI Chatbot Interfaces. arXiv:2604.06188
- Adarsh Kumarappan and Ananya Mijoo. 2026. Not Just RLHF: Why Alignment Alone Won’t Fix Multi-Agent Sycophancy. arXiv:2605.12991
- Veena Kumari and Pauldy C. J. Otermans. 2026. Chatbot psychosis: moving beyond recognition to mechanistic understanding and harm reduction. *The British Journal of Psychiatry* 229, 1 (2026), 1–3. doi:10.1192/bjp.2026.10541
- Jonathan Kutasov, Yuqi Sun, Paul Colognese, Teun van der Weij, Linda Petrini, Chen Bo Calvin Zhang, John Hughes, Xiang Deng, Henry Sleight, Tyler Tracy, et al. 2025. SHADE-Arena: Evaluating Sabotage and Monitoring in LLM Agents. arXiv:2506.15740
- Philippe Laban, Lidiya Murakhovska, Caiming Xiong, and Chien-Sheng Wu. 2023. Are You Sure? Challenging LLMs Leads to Performance Drops in The FlipFlop Experiment. arXiv:2311.08596
- Linnea Laestadius, Andrea Bishop, Michael Gonzalez, Diana Illeňčík, and Celeste Campos-Castillo. 2024. Too human and not human enough: A grounded theory analysis of mental health harms from emotional dependence on the social chatbot Replika. *New Media & Society* 26, 10 (2024), 5923–5941. doi:10.1177/14614448221142007
- Jixuan Leng, Chengsong Huang, Banghua Zhu, and Jiaxin Huang. 2025. Taming Overconfidence in LLMs: Reward Calibration in RLHF. In *The Thirteenth International Conference on Learning Representations (ICLR)*. 16484–16517.
- Benjamin A. Levinstein and Daniel A. Herrmann. 2025. Still no lie detector for language models: probing empirical and conceptual roadblocks. *Philosophical Studies* 182, 7 (2025), 1539–1565. doi:10.1007/s11098-023-02094-3
- Shuo Li, Tao Ji, Xiaoran Fan, Linsheng Lu, Leyi Yang, Yuming Yang, Zhiheng Xi, Rui Zheng, Yuran Wang, Xiaohui Zhao, et al. 2025a. Have the VLMs Lost Confidence? A Study of Sycophancy in VLMs. In *The Thirteenth International Conference on Learning Representations (ICLR)*. 2739–2759.
- Yubo Li, Ramayya Krishnan, and Rema Padman. 2026. Consistency of Large Reasoning Models Under Multi-Turn Attacks. arXiv:2602.13093

- Yubo Li, Xiaobin Shen, Yidi Miao, Xinyu Yao, Xueying Ding, Ramayya Krishnan, and Rema Padman. 2025b. Beyond Single-Turn: A Survey on Multi-Turn Interactions with Large Language Models. *arXiv:2504.04717*
- Joshua Liu, Aarav Jain, Soham Takuri, Srihan Vege, Aslihan Akalin, Kevin Zhu, Sean O'Brien, and Vasu Sharma. 2025a. TRUTH DECAY: Quantifying Multi-Turn Sycophancy in Language Models. *arXiv:2503.11656*
- Jiacheng Liu, Francesco Salvi, Philipp Schoenegger, Xiaoli Nan, Ramit Debnath, Barbara Fasolo, Evelina Leivada, Gabriel Recchia, Fritz Günther, Ali Zarifhonarvar, et al. 2026. When Large Language Models are More Persuasive Than Incentivized Humans, and Why. *arXiv:2505.09662*
- Minqian Liu, Zhiyang Xu, Xinyi Zhang, Heajun An, Sarvech Qadir, Qi Zhang, Pamela J. Wisniewski, Jin-Hee Cho, Sang Won Lee, Ruoxi Jia, et al. 2025b. LLM Can be a Dangerous Persuader: Empirical Study of Persuasion Safety in Large Language Models. In *Second Conference on Language Modeling (COLM 2025)*.
- Lars Malmqvist. 2025. Sycophancy in Large Language Models: Causes and Mitigations. In *Intelligent Computing (Lecture Notes in Networks and Systems)*, Kohei Arai (Ed.). Springer Nature Switzerland, 61–74. doi:10.1007/978-3-031-92611-2\_5
- Blazej Manczak, Eric Lin, Francisco Eiras, James O'Neill, and Vaikkunth Mugunthan. 2025. Shallow Robustness, Deep Vulnerabilities: Multi-Turn Evaluation of Medical LLMs. In *The Second Workshop on GenAI for Health: Potential, Trust, and Policy Compliance at NeurIPS 2025*. <https://arxiv.org/abs/2510.12255>
- Shuhaib Mehri, Xiaocheng Yang, Takyoun Kim, Gokhan Tur, Shikib Mehri, and Dilek Hakkani-Tür. 2026. Goal Alignment in LLM-Based User Simulators for Conversational AI. *Transactions of the Association for Computational Linguistics* 14 (2026), 852–876. doi:10.1162/tacl.a.687
- Ashish Mehta, Jared Moore, Jacy Reese Anthis, William Agnew, Eric Lin, Peggy Yin, Desmond C. Ong, Nick Haber, and Carol Dweck. 2026. The Dynamics of Delusion: Modeling Bidirectional False Belief Amplification in Human-Chatbot Dialogue. *arXiv:2604.25096*
- Miranda Muqing Miao and Lyle Ungar. 2026. Closing the Confidence-Faithfulness Gap in Large Language Models. *arXiv:2603.25052*
- Yifei Ming, Senthil Purushwalkam, Shrey Pandit, Zixuan Ke, Xuan-Phi Nguyen, Caiming Xiong, and Shafiq Joty. 2025. FaithEval: Can Your Language Model Stay Faithful to Context, Even If “The Moon is Made of Marshmallows”. In *The Thirteenth International Conference on Learning Representations (ICLR)*. 29430–29456.
- Parsa Mirtaheri and Mikhail Belkin. 2026. Catching rationalization in the act: detecting motivated reasoning before and after CoT via activation probing. In *Third Conference on Language Modeling (COLM 2026)*.
- Muhammad Ahmed Mohsin, Ahsan Bilal, Muhammad Umer, and Emily Fox. 2026. Pressure, What Pressure? Sycophancy Disentanglement in Language Models via Reward Decomposition. *arXiv:2604.05279*
- Luke Nicholls, Robert Hutto, Zephrah Soto, Hamilton Morrin, Thomas Pollak, Raj Korpan, and Cheryl Carmichael. 2026. “AI Psychosis” in Context: How Conversation History Shapes LLM Responses to Delusional Beliefs. *arXiv:2604.13860*
- Rodrigo Nogueira, Giovana Kerche Bonás, Thales Sales Almeida, Andrea Roque, Ramon Pires, Hugo Abonizio, Thiago Laitz, Celio Larcher, Roseval Malaquias Junior, and Marcos Piau. 2026. Measuring Opinion Bias and Sycophancy via LLM-based Persuasion. *arXiv:2604.21564*
- OpenAI. 2025. Expanding on what we missed with sycophancy. <https://openai.com/index/expanding-on-sycophancy/>.
- Hadas Orgad, Michael Tokar, Zorik Gekhman, Roi Reichart, Idan Szpektor, Hadas Kotek, and Yonatan Belinkov. 2025. LLMs Know More Than They Show: On the Intrinsic Representation of LLM Hallucinations. In *The Thirteenth International Conference on Learning Representations (ICLR)*. 66880–66913.
- Manav Pandey. 2026. LLMs Know They’re Wrong and Agree Anyway: The Shared Sycophancy-Lying Circuit. *arXiv:2604.19117*
- Henry Papadatos and Rachel Freedman. 2024. Linear Probe Penalties Reduce LLM Sycophancy. In *Socially Responsible Language Modelling Research (SoLaR) Workshop at NeurIPS 2024*. <https://arxiv.org/abs/2412.00967>
- Peter S. Park, Simon Goldstein, Aidan O’Gara, Michael Chen, and Dan Hendrycks. 2024. AI deception: A survey of examples, risks, and potential solutions. *Patterns* 5, 5 (2024), 100988. doi:10.1016/j.patter.2024.100988
- Dongshen Peng, Yi Wang, Austin Schoeffler, Sun-ha Hong, Brian Suffoletto, David Kim, Carl Preiksaitis, and Christian Rose. 2026. SycoEval-EM: Sycophancy Evaluation of Large Language Models in Simulated Clinical Encounters for Emergency Care. *arXiv:2601.16529*
- Ethan Perez, Sam Ringer, Kamilė Lukošaitė, Karina Nguyen, Edwin Chen, Scott Heiner, Craig Pettit, Catherine Olsson, Sandipan Kundu, Saurav Kadavath, et al. 2023. Discovering Language Model Behaviors with Model-Written Evaluations. In *Findings of the Association for Computational Linguistics: ACL 2023*. Association for Computational Linguistics, 13387–13434. doi:10.18653/v1/2023.findings-acl.847
- Renjie Pi, Kehao Miao, Peihang Li, Runtao Liu, Jiahui Gao, Jipeng Zhang, and Xiaofang Zhou. 2025. Pointing to a Llama and Call it a Camel: On the Sycophancy of Multimodal Large Language Models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 20166–20180. doi:10.18653/v1/2025.emnlp-main.1020

- Yuehan Qin, Shawn Li, Yi Nian, Xinyan Velocity Yu, Yue Zhao, and Xuezhe Ma. 2026. Don't Let It Hallucinate: Premise Verification via Retrieval-Augmented Logical Reasoning. *Transactions on Machine Learning Research* (2026). <https://openreview.net/forum?id=BDxStRGWba>
- Parisa Rabbani, Nimet Beyza Bozdag, and Dilek Hakkani-Tür. 2026. From Fact to Judgment: Investigating the Impact of Task Framing on LLM Conviction in Dialogue Systems. In *Proceedings of the 16th International Workshop on Spoken Dialogue System Technology*. Association for Computational Linguistics, 193–204. <https://aclanthology.org/2026.iwds-1.21/>
- Leonardo Ranaldi and Giulia Pucci. 2026. Learning Multilingual Agentic Policy to Control Sycophancy. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 3664–3681. doi:10.18653/v1/2026.eacl-long.169
- Richard Ren, Arunim Agarwal, Mantas Mazeika, Cristina Menghini, Robert Vacareanu, Brad Kenstler, Mick Yang, Isabelle Barrass, Alice Gatti, Xuwang Yin, et al. 2025. The MASK Benchmark: Disentangling Honesty From Accuracy in AI Systems. arXiv:2503.03750v3
- Mark Russinovich, Ahmed Salem, and Ronen Eldan. 2025. Great, Now Write an Article About That: The Crescendo Multi-Turn LLM Jailbreak Attack. In *34th USENIX Security Symposium (USENIX Security 25)*. USENIX Association, 2421–2440. <https://www.usenix.org/conference/usenixsecurity25/presentation/russinovich>
- Jeongwoo Ryu, Soomin Kim, Jinsu Eun, Kyusik Kim, Changhoon Oh, and Bongwon Suh. 2026. Feeling Right vs. Being Right: How AI Sycophancy Affects Value-Laden Deliberation. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 44227–44245. doi:10.18653/v1/2026.acl-long.2046
- Francesco Salvi, Manoel Horta Ribeiro, Riccardo Gallotti, and Robert West. 2025. On the conversational persuasiveness of GPT-4. *Nature Human Behaviour* 9, 8 (2025), 1645–1653. doi:10.1038/s41562-025-02194-6
- Francesco Salvi, Manoel Horta Ribeiro, Riccardo Gallotti, and Robert West. 2026. Author Correction: On the conversational persuasiveness of GPT-4. *Nature Human Behaviour* (2026). doi:10.1038/s41562-026-02588-0
- Ferdinand M. Schessl. 2026. The Autocorrelation Blind Spot: Why 42% of Turn-Level Findings in LLM Conversation Analysis May Be Spurious. arXiv:2604.14414
- Harshee Jignesh Shah. 2026. The Silicon Mirror: Dynamic Behavioral Gating for Anti-Sycophancy in LLM Agents. arXiv:2604.00478
- Itai Shapira, Gerdus Benade, and Ariel D. Procaccia. 2026. How RLHF Amplifies Sycophancy. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*.
- Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askill, Samuel R. Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R. Johnston, et al. 2024. Towards Understanding Sycophancy in Language Models. In *The Twelfth International Conference on Learning Representations (ICLR)*. 110–144.
- Soorya Ram Shingekar, Vipin Gunda, Jiwon Kim, Violeta J. Rodriguez, Hari Sundaram, and Koustuv Saha. 2026. AI Psychosis: Does Conversational AI Amplify Delusion-Related Language? arXiv:2603.19574
- Debu Sinha. 2026. SycoBench-600: Measuring Sycophancy and Correction Selectivity in LLM Assistants. In *Findings of the Association for Computational Linguistics: ACL 2026*. 35278–35284. doi:10.18653/v1/2026.findings-acl.1759
- Elias Stengel-Eskin, Peter Hase, and Mohit Bansal. 2025. Teaching Models to Balance Resisting and Accepting Persuasion. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Association for Computational Linguistics, 8108–8122. doi:10.18653/v1/2025.naacl-long.412
- Zhe Su, Xuhui Zhou, Sanketh Rangrejji, Anubha Kabra, Julia Mendelsohn, Faeze Brahman, and Maarten Sap. 2025. AI-LieDar: Examine the Trade-off Between Utility and Truthfulness in LLM Agents. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Association for Computational Linguistics, 11867–11894. doi:10.18653/v1/2025.naacl-long.595
- Xiangkun Sun, Ling kai Kong, Aoqi Zhang, Liang Zeng, and Tonghan Wang. 2026. How LLMs Are Persuaded: A Few Attention Heads, Rerouted. arXiv:2605.09314
- Yuan Sun and Ting Wang. 2026. Be Friendly, Not Friends: How LLM Sycophancy Shapes User Trust. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. Association for Computing Machinery, 1–15. doi:10.1145/3772318.3791079
- Bryan Chen Zhengyu Tan, Daniel Wai Kit Chin, Zhengyuan Liu, Nancy F. Chen, and Roy Ka-Wei Lee. 2025. Persuasion Dynamics in LLMs: Investigating Robustness and Adaptability in Knowledge and Safety with DuET-PD. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 1550–1575. doi:10.18653/v1/2025.emnlp-main.81
- H. Holden Thorp. 2026. Editorial Expression of Concern. *Science* 392, 6803 (2026), 1131. doi:10.1126/science.aej2383
- Nam Tran, Brianna Papoutsis, Mariana Ortiz, Angela Tran, Vincent Zhao, Nicole Neifert, Madeline Barrett, Stephanie Ko, Himani Devabhaktuni, Benjamin Nguyen, et al. 2026a. A Quantitative Analysis of a Reddit-based AI Psychosis Qualitative Dataset. PsyArXiv preprint. doi:10.31234/osf.io/7d6jw\_v1

- Nam Tran, Brianna Papoutsis, Mariana Ortiz, Angela Tran, Vincent Zhao, Nicole Neifert, Madeline Barrett, Stephanie Ko, Himani Devabhaktuni, Benjamin Nguyen, et al. 2026b. Digital Delusion: A Mixed Methods Exploration of AI Psychosis in Reddit Posts. *PsyArXiv preprint*. doi:10.31234/osf.io/ghpr7\_v1
- Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. 2023. Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. Curran Associates, Inc., 74952–74965. doi:10.52202/075280-3275
- Neeraj Varshney, Wenlin Yao, Hongming Zhang, Jianshu Chen, and Dong Yu. 2023. A Stitch in Time Saves Nine: Detecting and Mitigating Hallucinations of LLMs by Validating Low-Confidence Generation. *arXiv:2307.03987*
- Daniel Vennemeyer, Phan Anh Duong, Meryl Ye, Ruihong Huang, and Tianyu Jiang. 2026. Sycophantic Praise: Evaluating Excessive Praise in Language Models. *arXiv:2606.07441*
- Daniel Vennemeyer, Phan Anh Duong, Tiffany Zhan, and Tianyu Jiang. 2025. Sycophancy Is Not One Thing: Causal Separation of Sycophantic Behaviors in LLMs. *arXiv:2509.21305*
- Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, Chris Tar, Yun-Hsuan Sung, Denny Zhou, Quoc Le, et al. 2024. FreshLLMs: Refreshing Large Language Models with Search Engine Augmentation. In *Findings of the Association for Computational Linguistics: ACL 2024*. 13697–13720. doi:10.18653/v1/2024.findings-acl.813
- Keyu Wang, Jin Li, Shu Yang, Zhuoran Zhang, and Di Wang. 2026. When Truth Is Overridden: Uncovering the Internal Origins of Sycophancy in Large Language Models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 40. 33566–33574. doi:10.1609/aaai.v40i39.40645
- Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Lingpeng Kong, Qi Liu, Tianyu Liu, et al. 2024. Large Language Models are not Fair Evaluators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 9440–9450. doi:10.18653/v1/2024.acl-long.511
- Francis Rhys Ward, Francesco Belardinelli, Francesca Toni, and Tom Everitt. 2023. Honesty Is the Best Policy: Defining and Mitigating AI Deception. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. Curran Associates, Inc., 2313–2341. doi:10.52202/075280-0108
- Jerry Wei, Da Huang, Yifeng Lu, Denny Zhou, and Quoc V. Le. 2023. Simple Synthetic Data Reduces Sycophancy in Large Language Models. *arXiv:2308.03958*
- Jiaxin Wen, Ruiqi Zhong, Akbir Khan, Ethan Perez, Jacob Steinhardt, Minlie Huang, Samuel R. Bowman, He He, and Shi Feng. 2025. Language Models Learn to Mislead Humans via RLHF. In *The Thirteenth International Conference on Learning Representations (ICLR)*. 74670–74692.
- Marcus Williams, Micah Carroll, Adhyayan Narang, Constantin Weisser, Brendan Murphy, and Anca Dragan. 2025. On Targeted Manipulation and Deception when Optimizing LLMs for User Feedback. In *The Thirteenth International Conference on Learning Representations (ICLR)*. 89542–89593.
- Wenbin Wu. 2026. Aligning to Illusions: Choice Blindness in Human and AI Feedback. *arXiv:2603.08412*
- Zhengxuan Wu, Aryaman Arora, Atticus Geiger, Zheng Wang, Jing Huang, Dan Jurafsky, Christopher D. Manning, and Christopher Potts. 2025. AxBench: Steering LLMs? Even Simple Baselines Outperform Sparse Autoencoders. In *Proceedings of the 42nd International Conference on Machine Learning (ICML) (Proceedings of Machine Learning Research, Vol. 267)*. PMLR, 67035–67080.
- Zhishang Xiang, Zerui Chen, Yunbo Tang, Zhimin Wei, Ruqin Ning, Yujie Lin, Qinggang Zhang, and Jinsong Su. 2026. MemSyco-Bench: Benchmarking Sycophancy in Agent Memory. *arXiv:2607.01071*
- Boyu Xiao, Xiuqi Tian, Xuwen Song, Haochun Wang, Guanchun Song, Sendong Zhao, and Bing Qin. 2026. When Correct Beliefs Collapse: Epistemic Resilience of LLMs under Clinical Pressure. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 8720–8764. doi:10.18653/v1/2026.acl-long.395
- Jian Xie, Kai Zhang, Jiangjie Chen, Renze Lou, and Yu Su. 2024b. Adaptive Chameleon or Stubborn Sloth: Revealing the Behavior of Large Language Models in Knowledge Conflicts. In *The Twelfth International Conference on Learning Representations (ICLR)*. 35623–35646.
- Qiming Xie, Zengzhi Wang, Yi Feng, and Rui Xia. 2024a. Ask Again, Then Fail: Large Language Models' Vacillations in Judgment. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 10709–10745. doi:10.18653/v1/2024.acl-long.577
- Yizhe Xie, Congcong Zhu, Xinyue Zhang, Tianqing Zhu, Dayong Ye, Minfeng Qi, Huajie Chen, and Wanlei Zhou. 2026. From Spark to Fire: Modeling and Mitigating Error Cascades in LLM-Based Multi-Agent Collaboration. *arXiv:2603.04474*
- Hongshen Xu, Zichen Zhu, Situo Zhang, Da Ma, Shuai Fan, Lu Chen, and Kai Yu. 2024c. Rejection Improves Reliability: Training LLMs to Refuse Unknown Questions Using RL from Knowledge Feedback. In *First Conference on Language Modeling (COLM 2024)*.
- Rongwu Xu, Brian Lin, Shujian Yang, Tianqi Zhang, Weiyan Shi, Tianwei Zhang, Zhixuan Fang, Wei Xu, and Han Qiu. 2024a. The Earth is Flat because...: Investigating LLMs' Belief towards Misinformation via Persuasive Conversation.

- In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 16259–16303. doi:10.18653/v1/2024.acl-long.858
- Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. 2024b. Knowledge Conflicts for LLMs: A Survey. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 8541–8565. doi:10.18653/v1/2024.emnlp-main.486
- Xingcheng Xu, Jingjing Qu, Qiaosheng Zhang, Chaochao Lu, Yanqing Yang, Na Zou, and Xia Hu. 2026. Epistemic Traps: Rational Misalignment Driven by Model Misspecification. arXiv:2602.17676
- Ye Yang, Donghe Li, Zuchen Li, Fengyuan Li, Jingyi Liu, Li Sun, and Qingyu Yang. 2025. MisinfoBench: A Multi-Dimensional Benchmark for Evaluating LLMs' Resilience to Misinformation. In *Findings of the Association for Computational Linguistics: EMNLP 2025*. Association for Computational Linguistics, 10202–10229. doi:10.18653/v1/2025.findings-emnlp.540
- Binwei Yao, Chao Shang, Wanyu Du, Jianfeng He, Ruixue Lian, Yi Zhang, Hang Su, Sandesh Swamy, and Yanjun Qi. 2025a. Peacemaker or Troublemaker: How Sycophancy Shapes Multi-Agent Debate. arXiv:2509.23055
- Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. 2025b.  $\tau$ -bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains. In *The Thirteenth International Conference on Learning Representations (ICLR)*. 9965–10017.
- Meryl Ye, Lujain Ibrahim, Jessica Y. Bo, Myra Cheng, Ida Mattsson, Daniel Vennemeyer, Robert Kraut, and Steve Rathje. 2026. What Counts as AI Sycophancy? A Taxonomy and Expert Survey of a Fragmented Construct. arXiv:2605.21778
- Botai Yuan, Yutian Zhou, Yingjie Wang, Fushuo Huo, Yongcheng Jing, Li Shen, Ying Wei, Zhiqi Shen, Ziwei Liu, Tianwei Zhang, et al. 2025. EchoBench: Benchmarking Sycophancy in Medical Large Vision-Language Models. arXiv:2509.20146
- Hongbang Yuan, Pengfei Cao, Zhuoran Jin, Yubo Chen, Daojian Zeng, Kang Liu, and Jun Zhao. 2024. Whispers that Shake Foundations: Analyzing and Mitigating False Premise Hallucinations in Large Language Models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2670–2683. doi:10.18653/v1/2024.emnlp-main.155
- Hanning Zhang, Shizhe Diao, Yong Lin, Yi R. Fung, Qing Lian, Xingyao Wang, Yangyi Chen, Heng Ji, and Tong Zhang. 2024. R-Tuning: Instructing Large Language Models to Say 'I Don't Know'. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Association for Computational Linguistics, 7113–7139. doi:10.18653/v1/2024.naacl-long.394
- Kaiwei Zhang, Qi Jia, Zijian Chen, Wei Sun, Xiangyang Zhu, Chunyi Li, Dandan Zhu, and Guangtao Zhai. 2025. Sycophancy under Pressure: Evaluating and Mitigating Sycophantic Bias via Adversarial Dialogues in Scientific QA. arXiv:2508.13743
- Zhaohan Zhang, Chengzhengxu Li, Xiaoming Liu, Chao Shen, Ziquan Liu, and Ioannis Patras. 2026. Confidence Should Be Calibrated More Than One Turn Deep. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 38578–38594. doi:10.18653/v1/2026.acl-long.1787
- Jiale Zhao, Ke Fang, and Lu Cheng. 2026a. When and What to Ask: AskBench and Rubric-Guided RLVR for LLM Clarification. In *Findings of the Association for Computational Linguistics: ACL 2026*. Association for Computational Linguistics, 17120–17140. doi:10.18653/v1/2026.findings-acl.845
- Yunpu Zhao, Rui Zhang, Junbin Xiao, Changxin Ke, Ruibo Hou, Yifan Hao, and Ling Li. 2026b. Sycophancy in vision-language models: A systematic analysis and an inference-time mitigation framework. *Neurocomputing* 659 (2026), 131217. doi:10.1016/j.neucom.2025.131217
- Kaitlyn Zhou, Jena D. Hwang, Xiang Ren, Nouha Dziri, Dan Jurafsky, and Maarten Sap. 2025. REL-A.I.: An Interaction-Centered Approach To Measuring Human-LM Reliance. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Association for Computational Linguistics, 11148–11167. doi:10.18653/v1/2025.naacl-long.556
- Bin Zhu, Yinxuan Gui, Huiyan Qi, Jingjing Chen, Chong-Wah Ngo, and Ee-Peng Lim. 2026a. Benchmarking Gaslighting Negation Attacks Against Multimodal Large Language Models. In *Proceedings of the 2026 International Conference on Multimedia Retrieval (ICMR '26)*. Association for Computing Machinery, 2041–2049. doi:10.1145/3805622.3810624
- Bin Zhu, Hailong Yin, Jingjing Chen, and Yu-Gang Jiang. 2026b. Benchmarking Gaslighting Negation Attacks Against Reasoning Models. In *Multimedia Modeling (MMM 2026) (Lecture Notes in Computer Science)*. Springer Nature Singapore, 188–202. doi:10.1007/978-981-95-6960-1\_14
- Xiaochen Zhu, Caiqi Zhang, Tom Stafford, Nigel Collier, and Andreas Vlachos. 2025. Conformity in Large Language Models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 3854–3872. doi:10.18653/v1/2025.acl-long.195